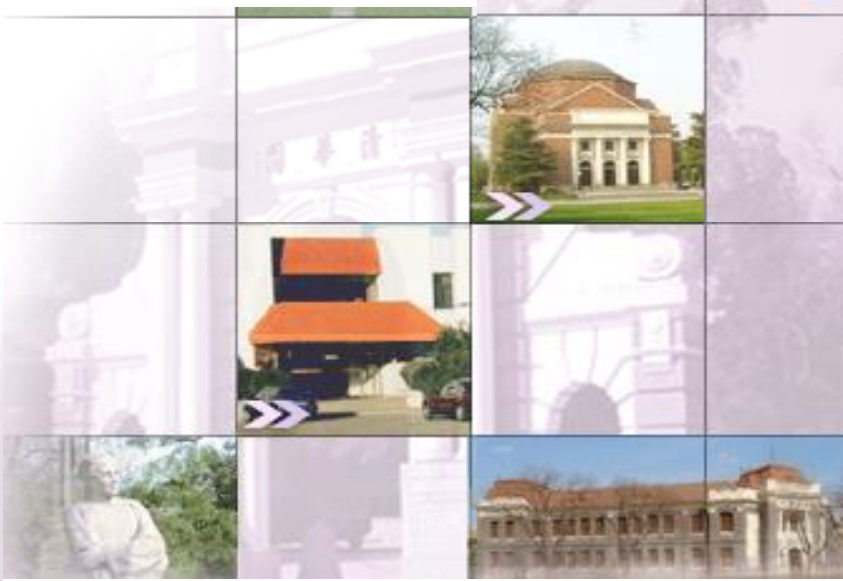
控制工程基础

（第五章）



5 控制系统稳定性分析

5.1系统稳定性的基本概念

5.2系统稳定的充要条件

5.3代数稳定性判据（Routh判据）

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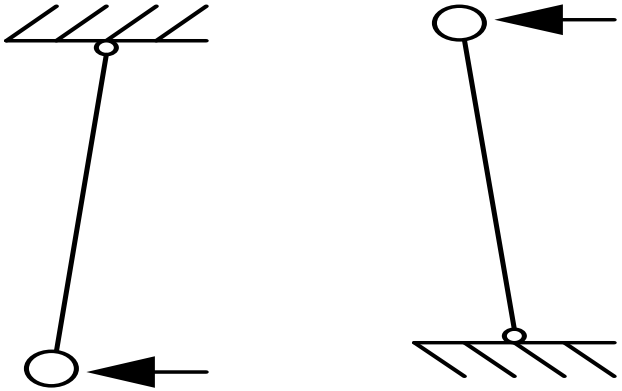
5.7控制系统的相对稳定性

5.1系统稳定性的基本概念

1. 单摆

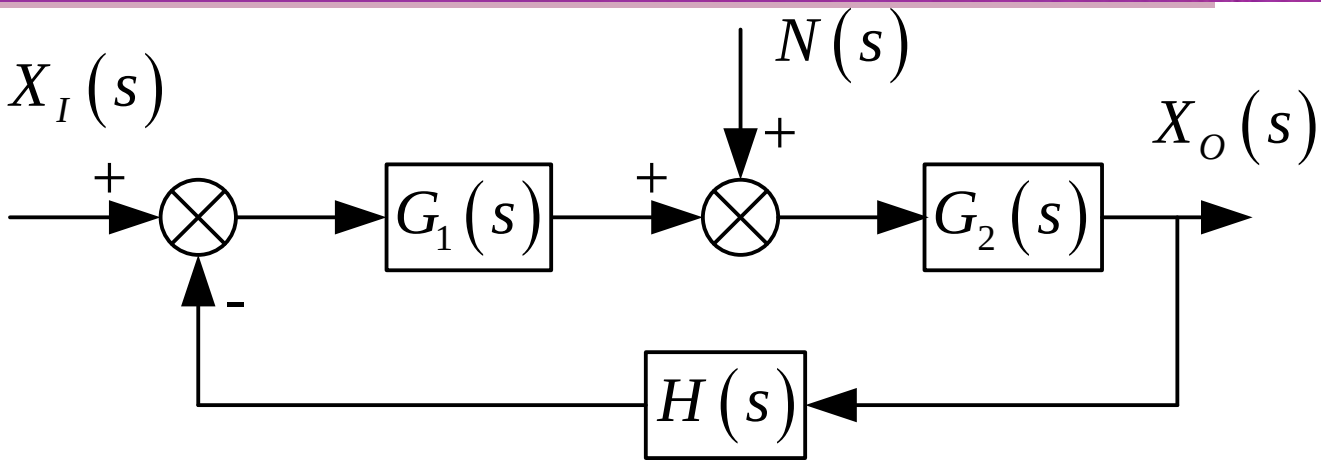


系统受扰动后能否恢复原来的状态

2. 闭环控制系统的稳定性问题

定义:控制系统在小初始偏差作用下，过渡过程逐渐衰减并趋于零，即具有恢复平衡状态的能力。

5.2 系统稳定的充要条件



N(s)到Xo(s)的传递函数：

$$\frac{X_o(s)}{N(s)}=\frac{G_2(s)}{1+G_1(s)G_2(s)H(s)}=\frac{b_0s^m+b_1s^{m-1}+\mathrm{L}\ +b_{m-1}s+b_m}{a_0s^n+a_1s^{n-1}+\mathrm{L}\ +a_{n-1}s+a_n}$$

设$n(t)$为单位脉冲函数， $N(s)=1$

$$X_O(s)=\frac{b_0s^m+b_1s^{m-1}+\mathrm{L}\ +b_{m-1}s+b_m}{a_0s^n+a_1s^{n-1}+\mathrm{L}\ +a_{n-1}s+a_n}$$

$$=\sum_i\frac{c_i}{s+\sigma_i}+\sum_j\left(\frac{d_j}{s^2+2\zeta_j\omega_js+\omega_j^2}+\frac{e_js}{s^2+2\zeta_j\omega_js+\omega_j^2}\right)$$

$$\frac{1}{s+\sigma} \rightarrow \quad e^{-\sigma t}$$

$$\frac{1}{s^2+2\zeta\omega s+\omega^2} \rightarrow \quad \frac{1}{\omega\sqrt{1-\zeta^2}} e^{-\zeta\omega t} \sin \omega\sqrt{1-\zeta^2}\, t$$

二阶欠阻尼系统单位脉冲响应

$$\frac{s}{s^2+2\zeta\omega s+\omega^2} \rightarrow$$

$$\frac{-1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega t} \sin\left(\omega\sqrt{1-\zeta^2}\, t - \arctan\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$

二阶欠阻尼系统单位阶跃响应

$$x_o(t)=\sum_i f_i e^{-\sigma_i t}+$$

$$\sum_j g_j e^{-\zeta_j \omega_j t}\sin\left(\omega_j\sqrt{1-\zeta_j^2}t+\varphi_j\right)$$

如果系统稳定，应有 $\underset{t\to\infty}{x_o(t)}=0$

即 $-\sigma_i<0,\ -\zeta_j\omega_j<0$

$$-\sigma_i < 0, \quad -\zeta_j \omega_j < 0$$

$$\frac{1}{s+\sigma}, \; s+\sigma = 0\text{的根}: s = -\sigma$$

$$\frac{1}{s^2+2\zeta\omega s+\omega^2}, \; s^2+2\zeta\omega s+\omega^2 = 0\text{的根}:$$

$$\frac{-2\zeta\omega \pm \sqrt{4\zeta^2\omega^2 - 4\omega^2}}{2} = -\zeta\omega \pm j\omega\sqrt{1-\zeta^2}$$

$-\sigma, -\zeta\omega$ 为系统闭环特征方程式的根的实部

控制系统稳定的充分必要条件是：
闭环特征方程式的根全部具有负实部

系统特征根即闭环极点，
故也可以说充要条件为
极点全部在[s]平面的左半面

$$x^3 + px + q = 0$$

$$x_1 = \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}}$$

五次以及更高次的代数方程没有一般的代数解法（即由方程的系数经有限次四则运算和开方运算求根的方法）

——阿贝尔定理

基于方程式的根与系数的关系

设系统特征方程为

$$a_0 s^n + a_1 s^{n-1} + \mathrm{L} + a_{n-1} s + a_n = 0$$

$$a_0 \left(s^n + \frac{a_1}{a_0} s^{n-1} + \mathrm{L} + \frac{a_{n-1}}{a_0} s + \frac{a_n}{a_0} \right) = 0$$

$$a_0 (s - s_1)(s - s_2) \mathrm{L} (s - s_n) = 0$$

$s_1, s_2, \mathrm{L}, s_n$ 为系统的特征根

复数根与系数的关系：

$$
\begin{cases}
\dfrac{a_1}{a_0} = -\left(s_1 + s_2 + \cdots + s_n\right); \\
\dfrac{a_2}{a_0} = +\left(s_1 s_2 + s_1 s_3 + \cdots + s_{n-1} s_n\right); \\
\dfrac{a_3}{a_0} = -\left(s_1 s_2 s_3 + s_1 s_2 s_4 + \cdots + s_{n-2} s_{n-1} s_n\right); \\
\cdots\cdots\cdots\cdots \\
\dfrac{a_n}{a_0} = \left(-1\right)^n \left(s_1 s_2 s_3 \cdots s_{n-2} s_{n-1} s_n\right)
\end{cases}
$$

要使全部特征根均具有负实部，必须满足：

（1）特征方程的各项系数 $a_i \neq 0$
(*i*=0，1，2，…，n)。

（2）特征方程的各项系数的符号都相同。

a_i 一般取正值，则上述两条件简化为 $a_i > 0$

——必要条件！

充要条件：

如果“劳斯阵列”中第一列所有项均为正，则系统稳定。

劳斯阵列：

$$
\begin{array}{cccccc}
s^n & a_0 & a_2 & a_4 & a_6 & \mathrm{L} \\
s^{n-1} & a_1 & a_3 & a_5 & a_7 & \mathrm{L} \\
s^{n-2} & b_1 & b_2 & b_3 & b_4 & \mathrm{L} \\
s^{n-3} & c_1 & c_2 & c_3 & c_4 & \mathrm{L} \\
\mathrm{M} & \mathrm{M} & \mathrm{M} & \mathrm{M} & & \\
s^2 & u_1 & u_2 & & & \\
s^1 & v_1 & & & & \\
s^0 & w_1 & & & &
\end{array}
$$

其中

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

$$b_3 = \frac{a_1 a_6 - a_0 a_7}{a_1}$$

$$\cdots\cdots$$

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}$$

$$c_2 = \frac{b_1 a_5 - a_1 b_3}{b_1}$$

$$c_3 = \frac{b_1 a_7 - a_1 b_4}{b_1}$$

$$\cdots\cdots$$

$$d_1 = \frac{c_1 b_2 - b_1 c_2}{c_1}$$

实部为正的特征根数＝

劳斯阵列中第一列的系数符号改变的次数。

例：　设控制系统的特征方程式为

$$s^4+8s^3+17s^2+16s+5=0$$

试应用劳斯稳定判据判断系统的稳定性。

解：$s^4+8s^3+17s^2+16s+5=0$

首先由方程系数可知满足稳定的必要条件。

其次，排劳斯阵列

s^4	1	17	5
s^3	8	16	
s^2	15	5	
s^1	$\frac{40}{3}$		
s^0	5		

劳斯阵列第一列中系数符号全为正，所以控制系统稳定。

例2 设控制系统的特征方程式为

$$s^4+2s^3+3s^2+4s+3=0$$

试应用劳斯稳定判据判断系统的稳定性。

解：由方程系数可知已满足稳定的必要条件。 排劳斯阵列

s^4	1	3	3
s^3	2	4	
s^2	1	3	
s^1	-2		
s^0	3		

第一列系数改变符号2次，闭环系统的根中有2个实部为正，控制系统不稳定。

对于特征方程阶次低（n≤3）的系统，劳斯判据可简化：

二阶系统特征式为 $a_0 s^2 + a_1 s + a_2$ ，劳斯表为

s^2	a_0	a_2
s^1	a_1	
s^0	a_2	

故二阶系统稳定的充要条件是

$$a_0 > 0, a_1 > 0, a_2 > 0$$

三阶系统特征式为 $a_0s^3+a_1s^2+a_2s+a_3$ ，

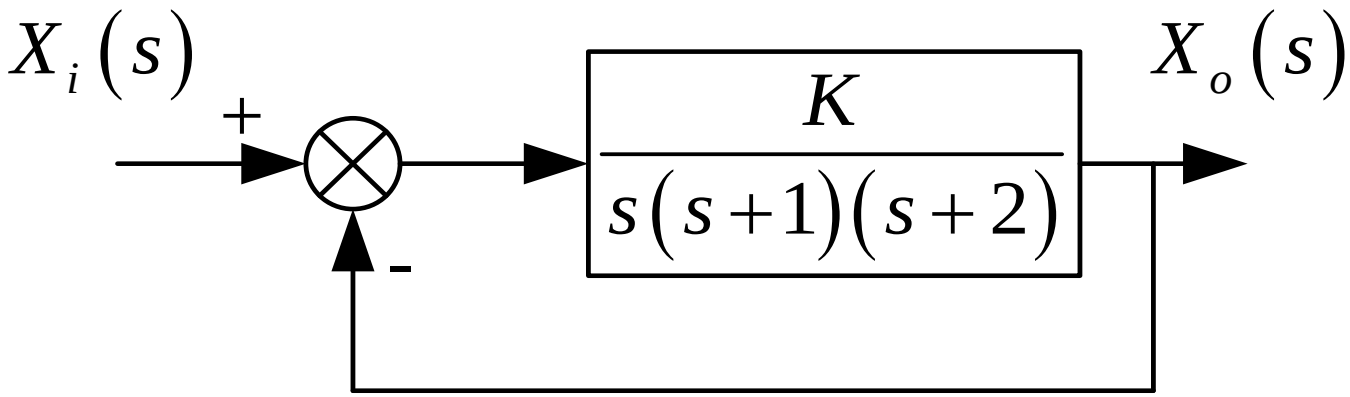
劳斯表:

$$
\begin{array}{lll}
s^3 & a_0 & a_2 \\
s^2 & a_1 & a_3 \\
s^1 & \dfrac{a_1a_2-a_0a_3}{a_1} & \\
s^0 & a_3 &
\end{array}
$$

故三阶系统稳定的充要条件是

$$a_0, a_1, a_2, a_3 > 0, a_1a_2 > a_0a_3$$

例 设某反馈控制系统如下图所示，试计算使系统稳定的K值范围。



解：系统闭环传递函数为

$$\frac{X_o(s)}{X_i(s)}=\frac{K}{s(s+1)(s+2)+K}$$

特征方程为

$$s(s+1)(s+2)+K=s^3+3s^2+2s+K=0$$

根据**三阶系统稳定的充要条件**，
可知使系统稳定须满足

$$\begin{cases} K>0 \\ 2\times 3>K\times 1 \end{cases}$$

故使系统稳定的K值范围为 $0<K<6$

例：　设控制系统的闭环特征方程式为

$$s^4+2s^3+s^2+2s+1=0$$

用劳斯判据判断稳定性。

劳斯阵列表

s^4	1	1	1
s^3	2	2	
s^2	$0(\varepsilon)$	1	
s^1	$2-\dfrac{2}{\varepsilon}$		
s^0	1		

符号改变2次，
2个正实根。

$$s^3 + 2s^2 + s + 2 = 0$$

s^3	1	1
s^2	2	2
s^1	$0(\varepsilon)$	
s^0	2	

无正实根，
有虚根。

例：　设控制系统的闭环特征方程式为

$$s^6+2s^5+8s^4+12s^3+20s^2+16s+16=0$$

用劳斯判据判断稳定性。

劳斯阵列表

s^6	1	8	20	16
s^5	2	12	16	0
s^4	1	6	8	
s^3	0	0	0	
	4	12		
s^2	3	8		
s^1	$\frac{4}{3}$			
s^0	8			

$$A(s)=s^4+6s^2+8$$

$$\frac{dA(s)}{ds}=4s^3+12s$$

临界稳定

赫尔维茨稳定性判据

$$a_0 s^n + a_1 s^{n-1} + \mathrm{L} + a_{n-1} s + a_n = 0, \qquad a_0 > 0$$

n×n行列式：

$$\Delta = \begin{vmatrix} a_1 & a_3 & a_5 & \mathrm{L} & & & 0 \\ a_0 & a_2 & a_4 & \mathrm{L} & & & \\ \mathrm{M} & a_1 & a_3 & a_5 & \mathrm{L} & & \\ & a_0 & a_2 & a_4 & \mathrm{L} & & \\ & \mathrm{M} & \mathrm{M} & \mathrm{O} & & & \\ & & & & & a_{n-1} & 0 \\ 0 & & & & & a_{n-2} & a_n \end{vmatrix}$$

系统稳定的充要条件：

各阶主子行列式均 >0

即：

$$\Delta_1 = a_1 > 0 \qquad \Delta_2 = \begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix} > 0$$

$$\Delta_3 = \begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} > 0 \qquad \cdots\cdots$$

例：设控制系统的特征方程式为

$$s^4+8s^3+17s^2+16s+5=0$$

试应用赫尔维茨稳定判据判断系统的稳定性。

解：由方程系数可知满足稳定的必要条件。

各系数排成行列式

$$\Delta=\begin{vmatrix} 8 & 16 & 0 & 0 \\ 1 & 17 & 5 & 0 \\ 0 & 8 & 16 & 0 \\ 0 & 1 & 17 & 5 \end{vmatrix}$$

由于

$$\Delta_1 = 8 > 0$$

$$\Delta_2 = \begin{vmatrix} 8 & 16 \\ 1 & 17 \end{vmatrix} > 0$$

$$\Delta_3 = \begin{vmatrix} 8 & 16 & 0 \\ 1 & 17 & 5 \\ 0 & 8 & 16 \end{vmatrix} > 0$$

$$\Delta = \begin{vmatrix} 8 & 16 & 0 & 0 \\ 1 & 17 & 5 & 0 \\ 0 & 8 & 16 & 0 \\ 0 & 1 & 17 & 5 \end{vmatrix} > 0$$

故该系统稳定。

劳斯判据和赫尔维茨判据均是通过特征根与系数的关系进行稳定性判断，判别式均为代数式，属于代数稳定性判据。

代数稳定性判据使用的多项式

是系统闭环特征多项式。

代数稳定性判据的不足：

- 必须知道系统的闭环传递函数
- 定性——较难从量上判断系统的稳定程度
- 对含有延迟环节的系统无效

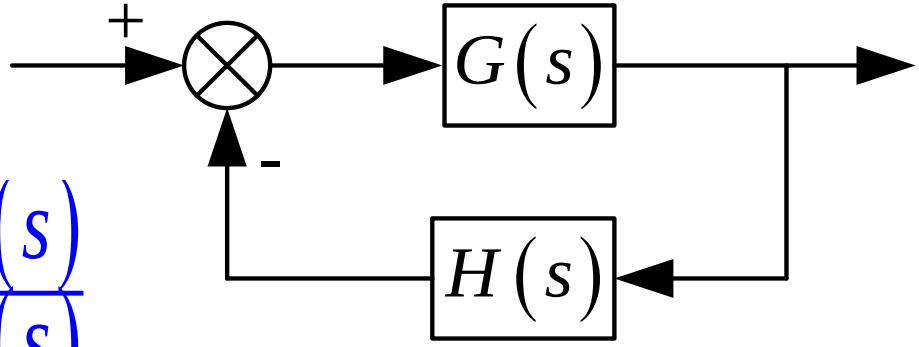
几何稳定性判据（Nyquist稳定判据、Bode稳定判据）

根据**开环频率特性**判断闭环稳定性

5.4乃奎斯特稳定性判据

控制系统的稳定性是指闭环极点全部位于[s]平面的负半平面，而奈奎斯特稳定性判据是利用开环传递函数判断闭环系统的稳定性。因此，需要搞清楚开环零极点和闭环零极点之间的关系。

反馈控制系统



$$G(s)=\frac{B_1(s)}{A_1(s)},\quad H(s)=\frac{B_2(s)}{A_2(s)}$$

开环传递函数

$$G(s)H(s)=\frac{B_1(s)B_2(s)}{A_1(s)A_2(s)}$$

闭环传递函数

$$\frac{G(s)}{1+G(s)H(s)}=\frac{\dfrac{B_1(s)}{A_1(s)}}{1+\dfrac{B_1(s)B_2(s)}{A_1(s)A_2(s)}}=\frac{B_1(s)A_2(s)}{A_1(s)A_2(s)+B_1(s)B_2(s)}$$

闭环稳定⇔闭环传递函数右极点个数为0

$$\frac{B_1(s)A_2(s)}{A_1(s)A_2(s)+B_1(s)B_2(s)}$$

⇔ $A_1(s)A_2(s)+B_1(s)B_2(s)$右零点个数为0

辅助传递函数$\boldsymbol{F(s)=1+G(s)H(s)}$

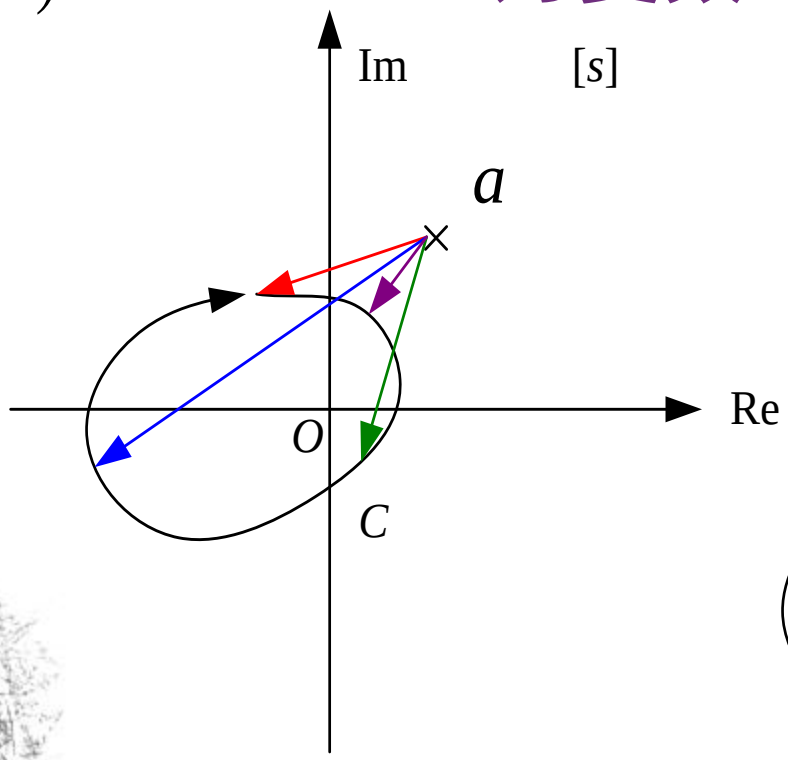
$$\frac{A_1(s)A_2(s)+B_1(s)B_2(s)}{A_1(s)A_2(s)}=1+\frac{B_1(s)B_2(s)}{A_1(s)A_2(s)}=1+G(s)H(s)$$

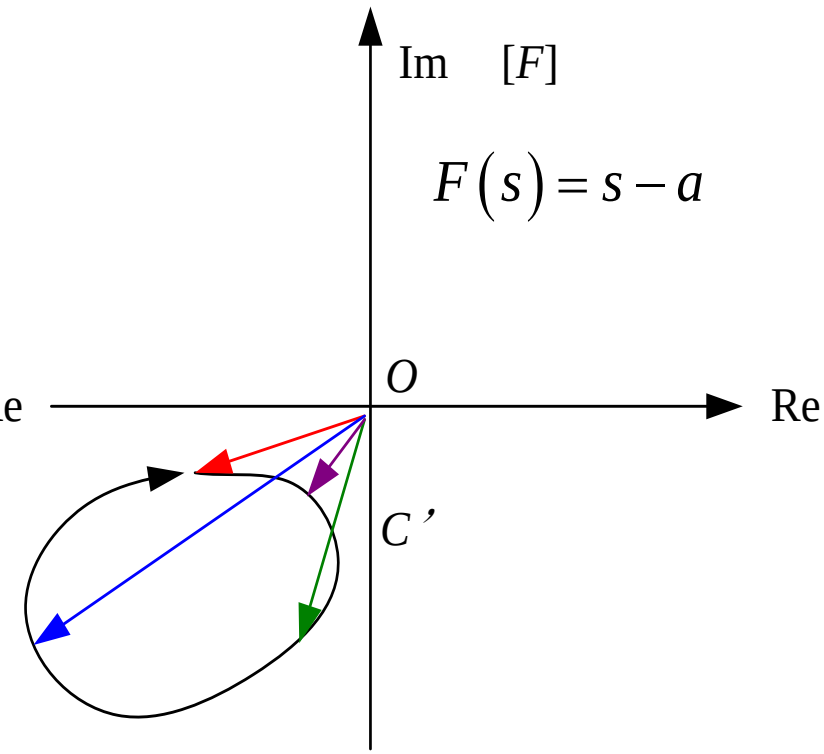
开环传递函数的极点为辅助传递函数的极点

辅助传递函数的零点为闭环传递函数的极点

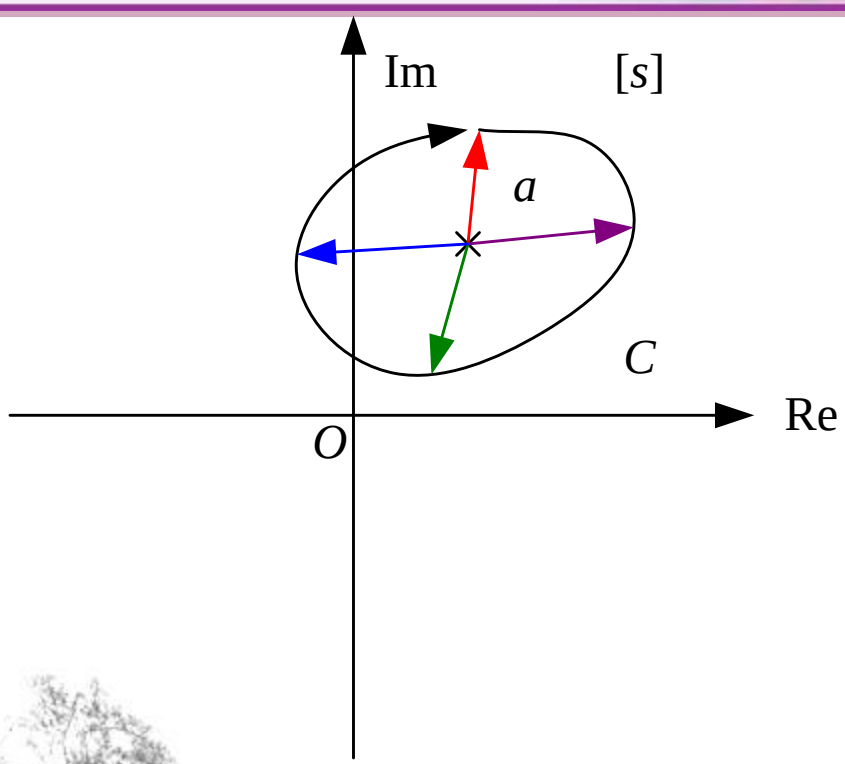
$s \to F(s)$

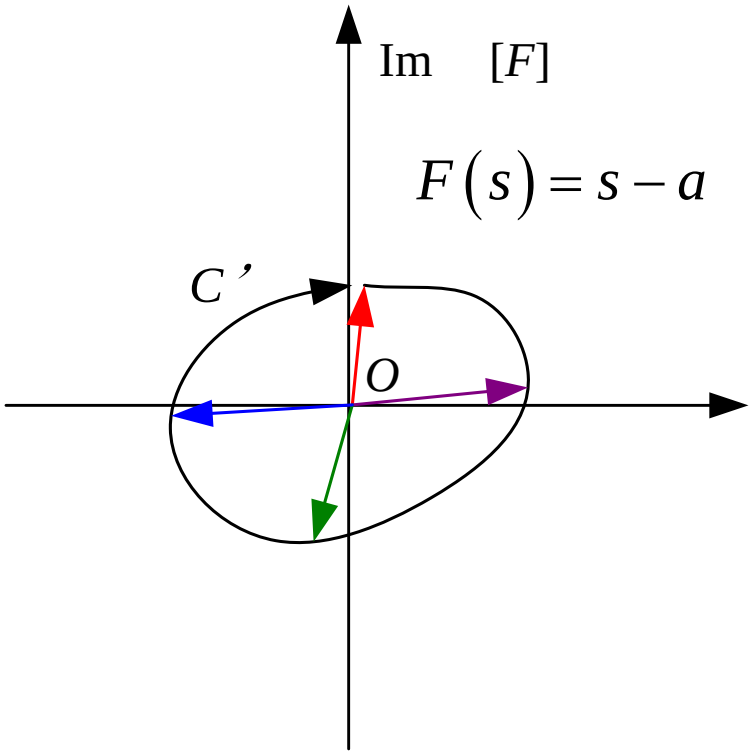
1. $F(s) = s - a$ a 为复数





C 为顺时针方向

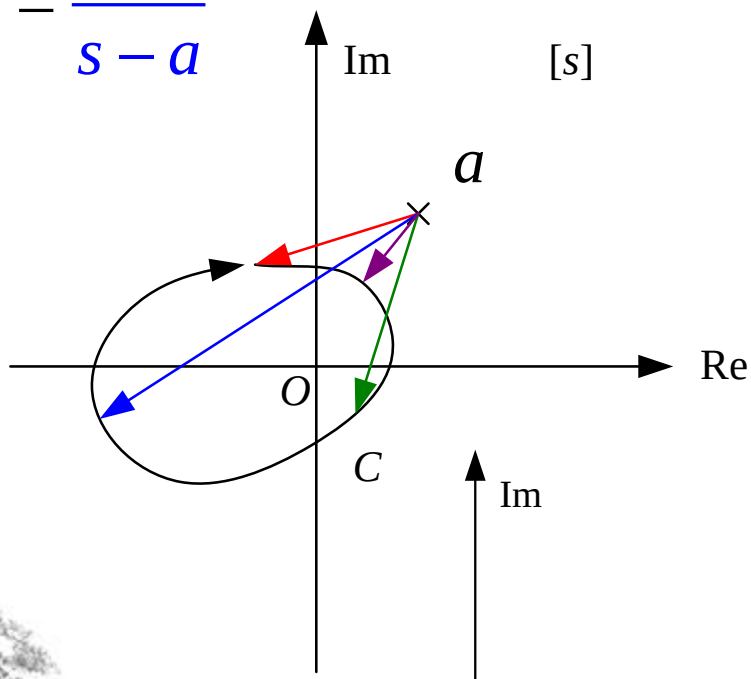


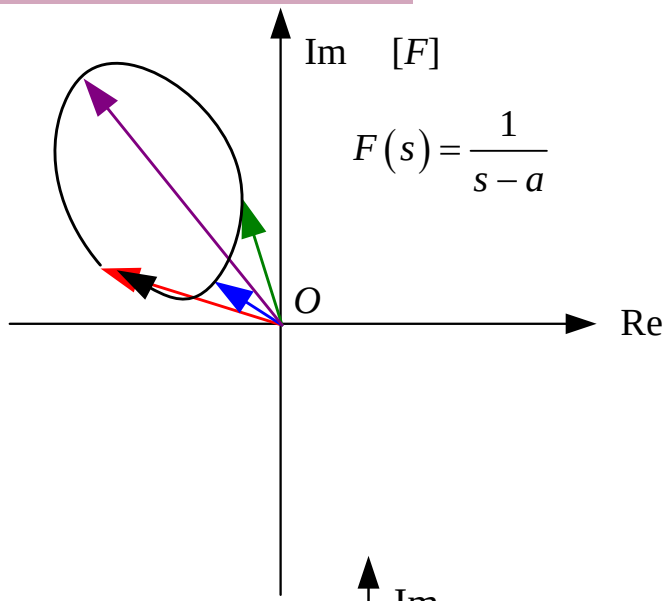


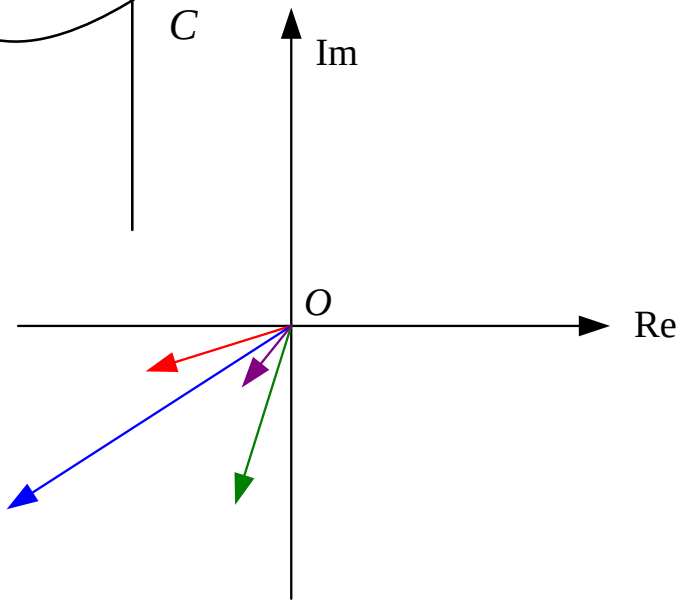
如果 C 包围 a，则 C' 顺时针包围原点1圈；

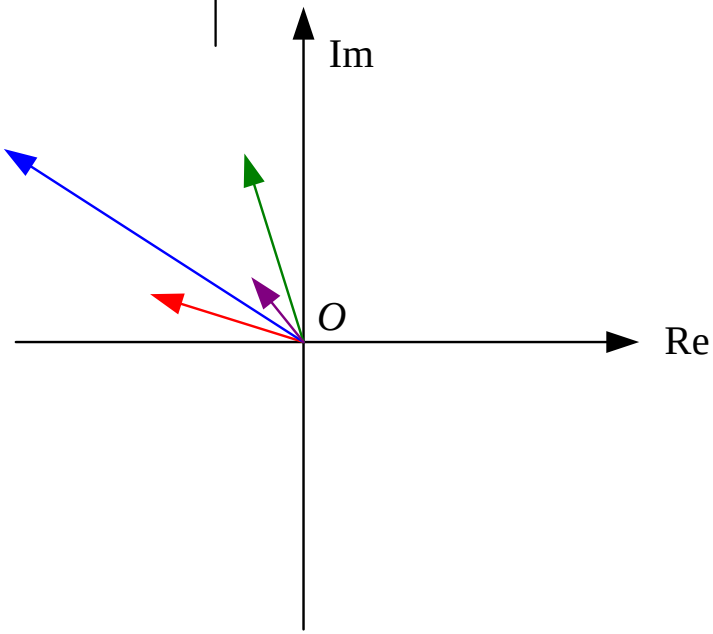
如果 C 不包围 a，则C' 不包围原点。

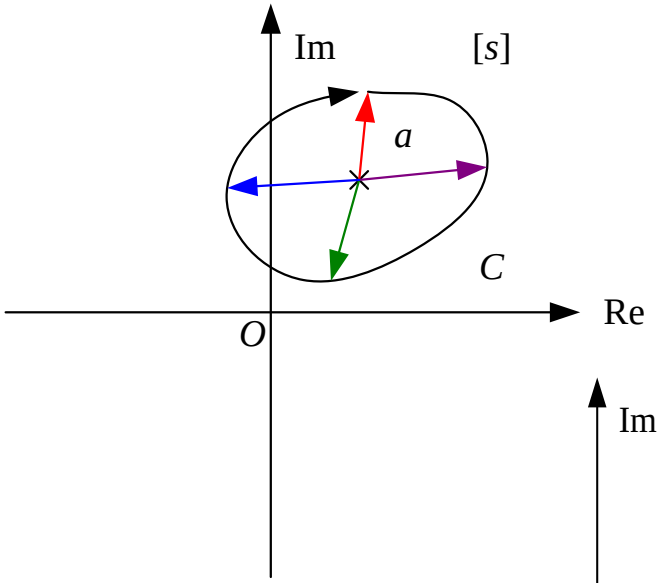
2. $F(s)=\dfrac{1}{s-a}$



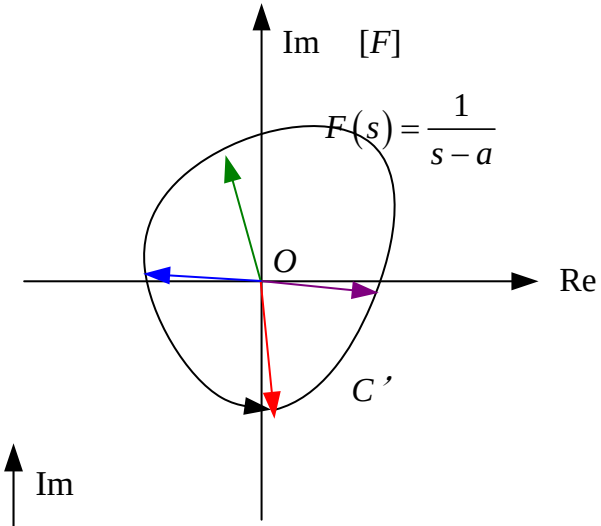


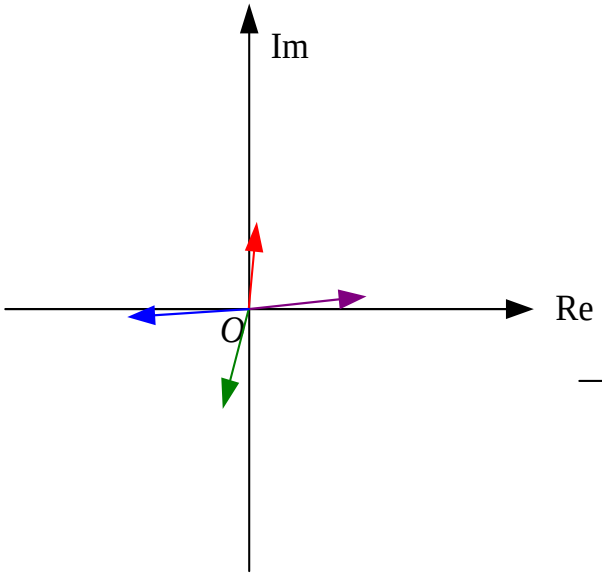


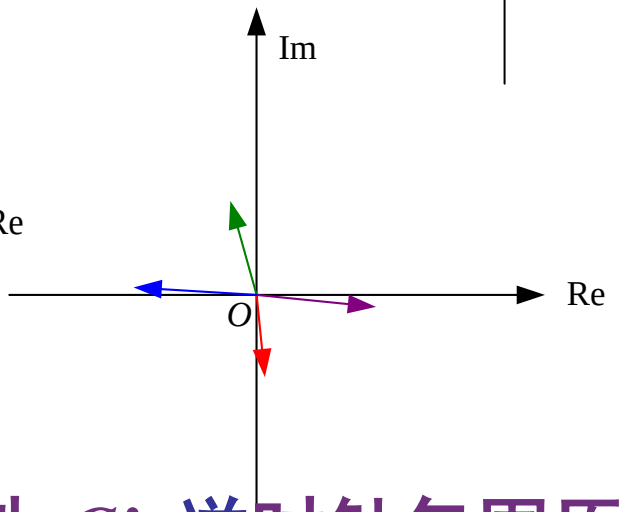




$$F(s)=\frac{1}{s-a}$$



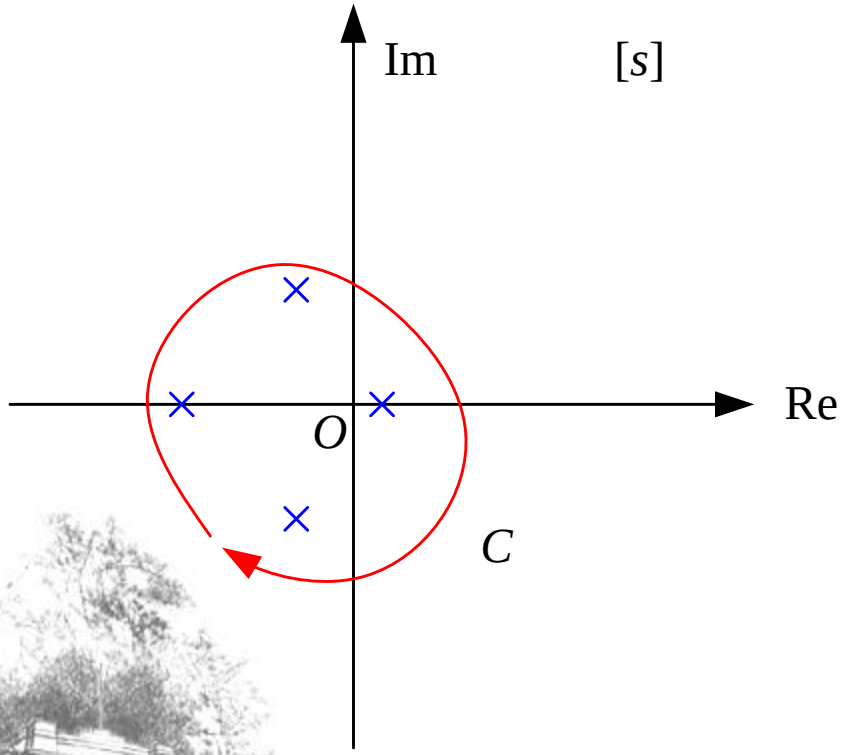


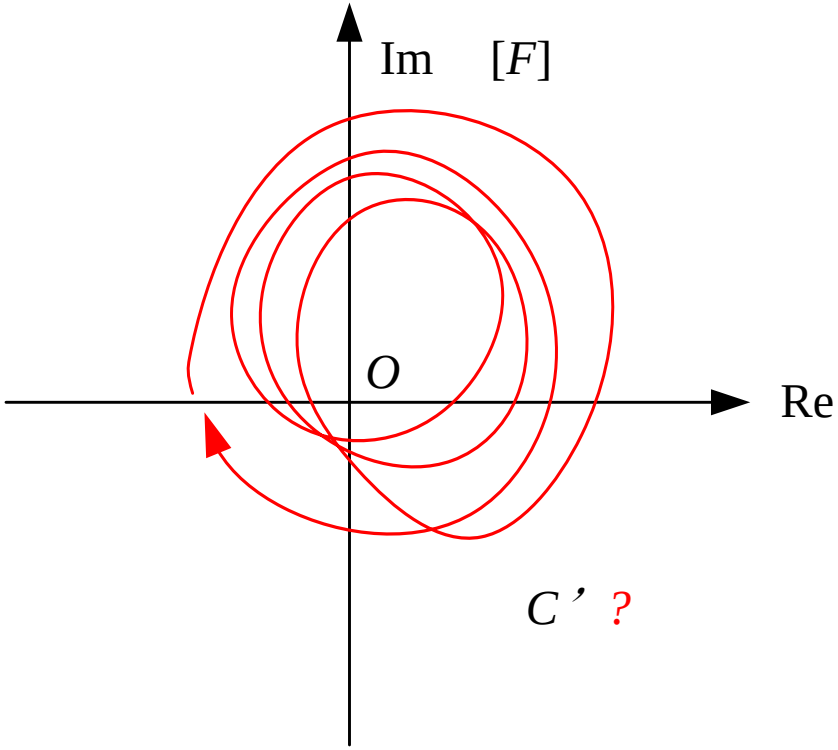


如果 C 包围 a，则 C' 逆时针包围原点1圈；

如果 C 不包围 a，则C' 不包围原点。

$$F(s)=(s-a_1)(s-a_2)\mathrm{L}\ (s-a_z)$$

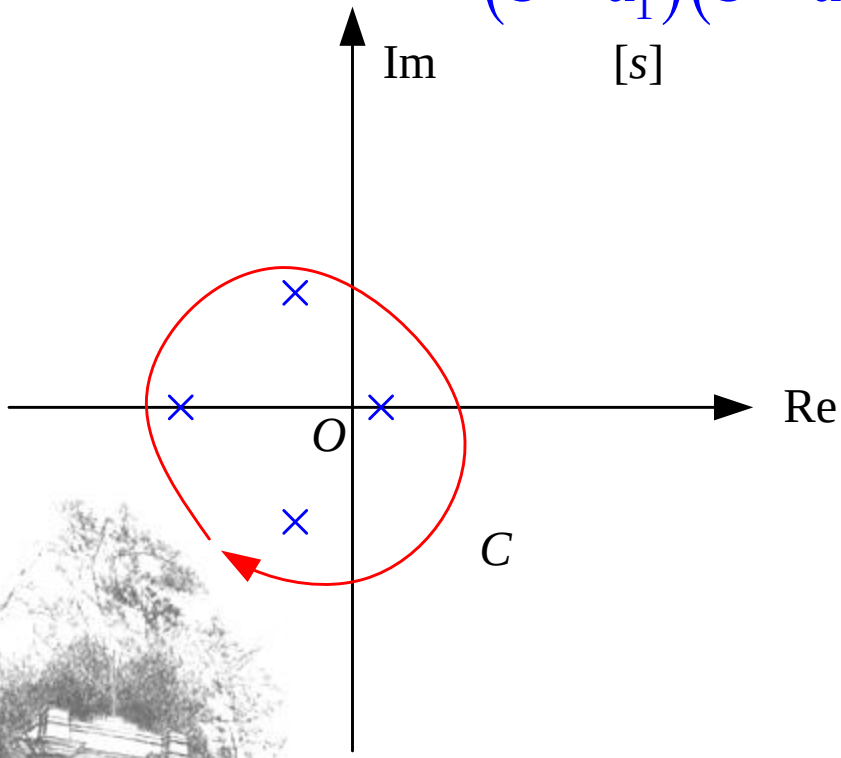


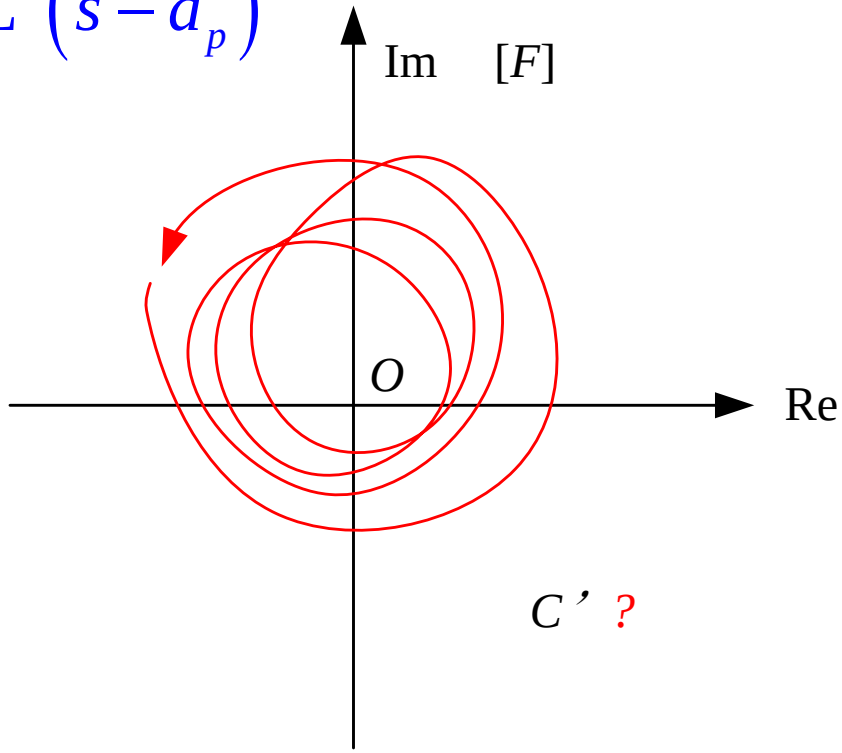


顺时针绕原点1圈，角度增量 -2π

***C*包围*z*个零点，*C*’绕原点 顺时针*z*圈**

$$F(s)=\frac{1}{(s-a_1)(s-a_2)\mathrm{L}\,(s-a_p)}$$





C包围1个极点，C'逆时针绕原点1圈

C包围p个极点，C'绕原点 逆时针p圈

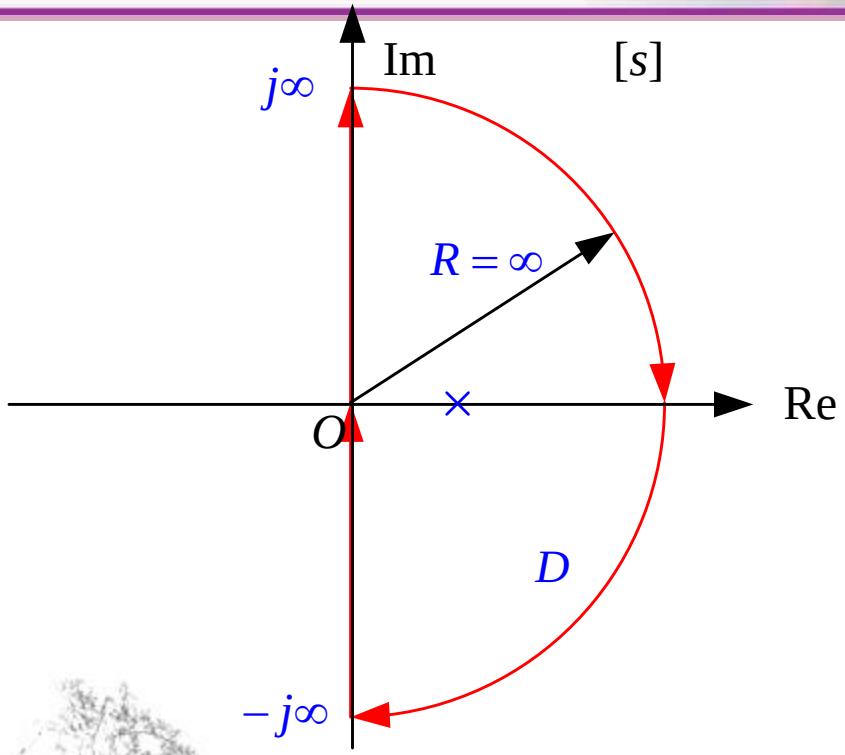
$$F(s)=\frac{(s-a_1)(s-a_2)\mathrm{L}\ (s-a_m)}{(s-a_1)(s-a_2)\mathrm{L}\ (s-a_n)}$$

$F(s)$有m个零点，n个极点，

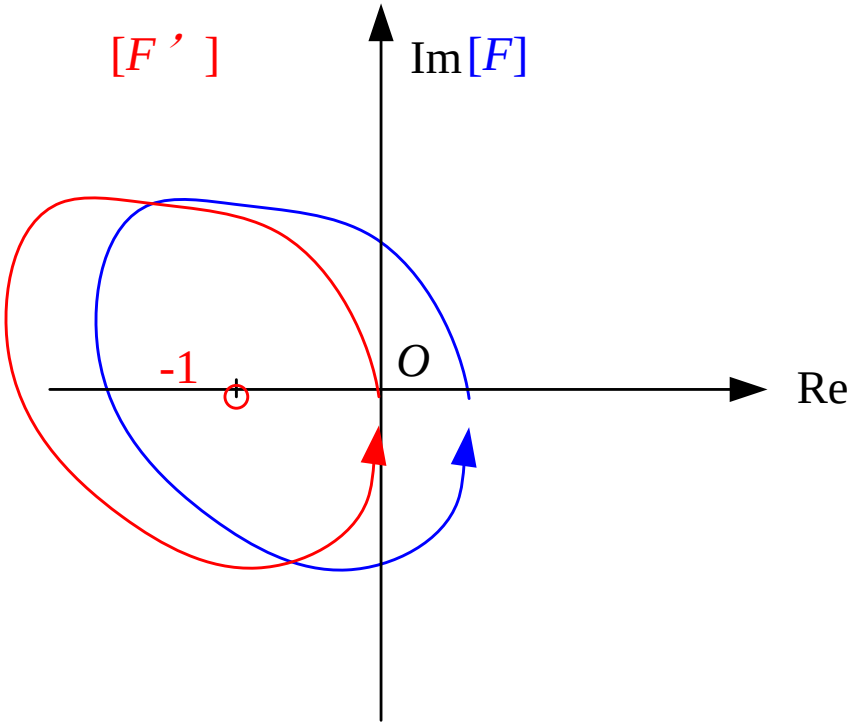
在$[s]$平面上的封闭曲线C顺时针如果包围了其中z个零点和p个极点，

则经过$F(s)$映射以后，在$[F]$平面上的C'曲线顺时针包围原点$z-p$圈。

——映射定理



$$F(s)=1+G(s)H(s)$$



$$F'(s)=G(s)H(s)$$

$F(s)$包围原点的圈数 = $F'(s)$包围-1点的圈数

Nyquist稳定判据

在[s]平面作包围右半平面的*D*形曲线，
如果开环传递函数的Nyquist图逆时针包围
(－1, j0)点的圈数等于开环右极点的个数，
则系统稳定。

——充要条件

闭环稳定⇔闭环传递函数右极点个数为0

$$\frac{B_1(s)A_2(s)}{A_1(s)A_2(s)+B_1(s)B_2(s)}$$

⇔ $A_1(s)A_2(s)+B_1(s)B_2(s)$右零点个数为0

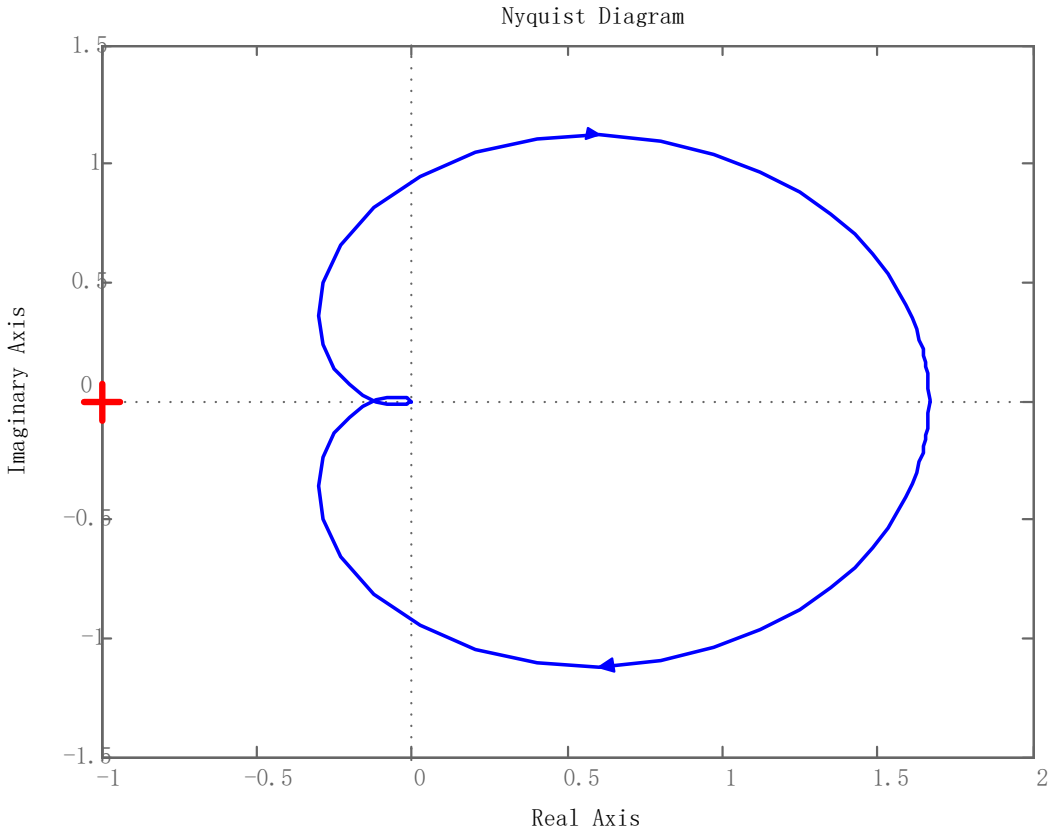
$$\frac{A_1(s)A_2(s)+B_1(s)B_2(s)}{A_1(s)A_2(s)}=1+\frac{B_1(s)B_2(s)}{A_1(s)A_2(s)}=1+G(s)H(s)$$

⇔顺时针绕[s]右半平面的曲线，经过

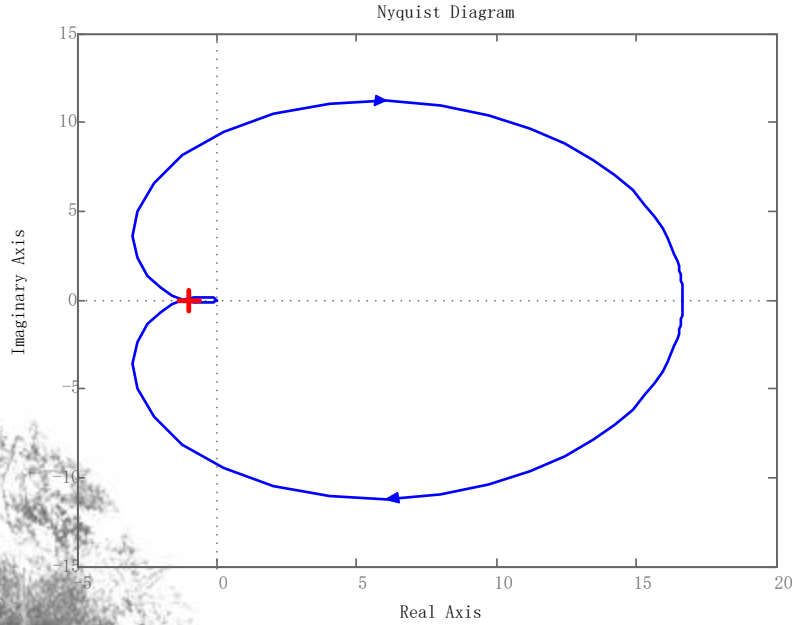
$F(s)=1+G(s)H(s)$ 的映射，

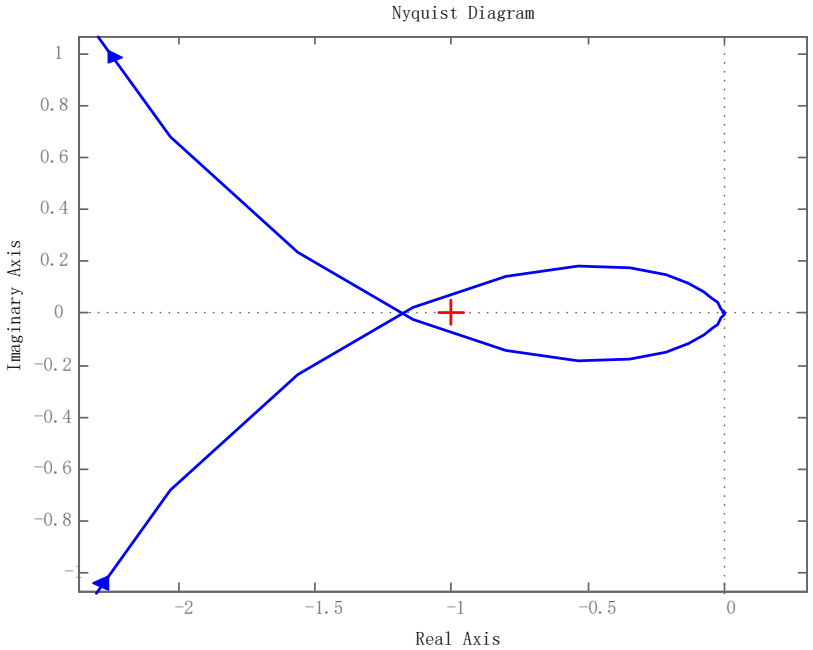
逆时针包围原点的圈数 = 开环右极点个数

$$G(s)H(s)=\frac{K}{(s+1)(s+2)(s+6)},\ K=20$$

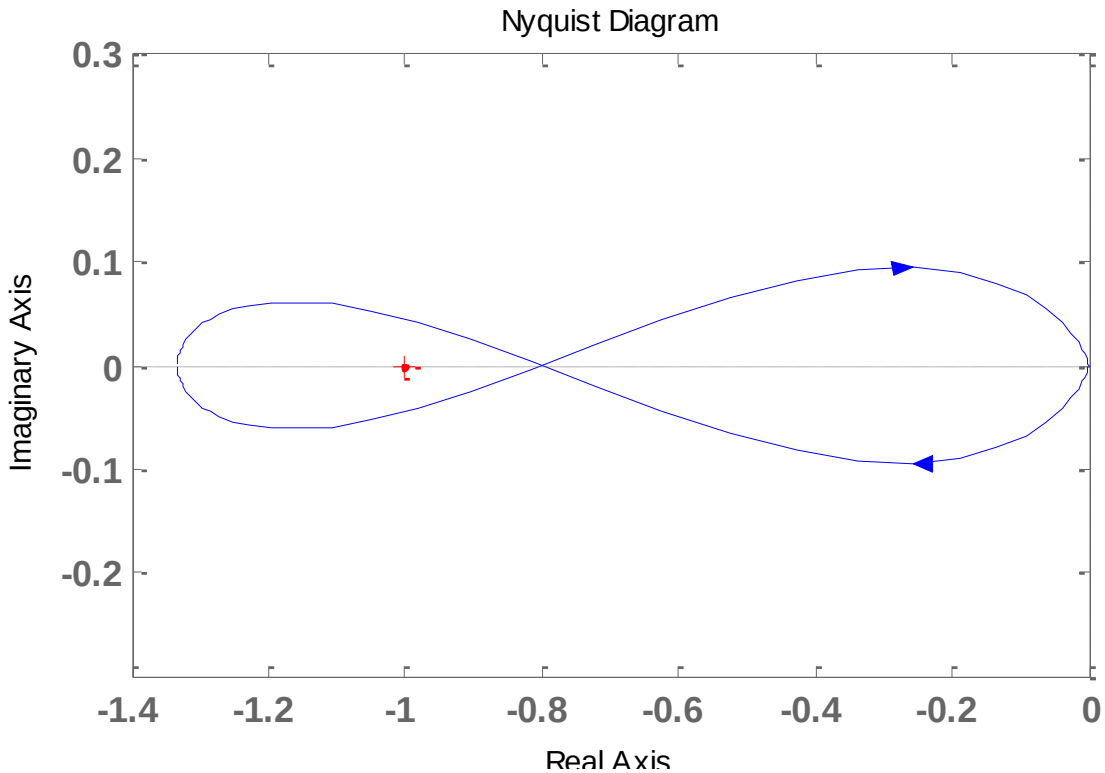


$$G(s)H(s)=\frac{K}{(s+1)(s+2)(s+6)},\ K=200$$





$$G(s)H(s)=\frac{8}{(s-1)(s+2)(s+3)}$$

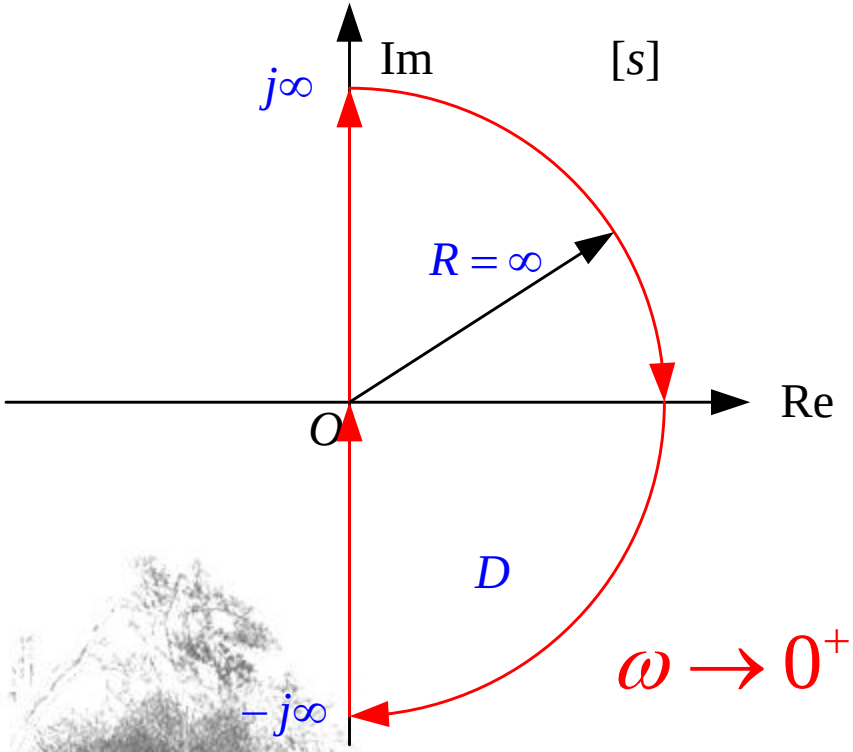


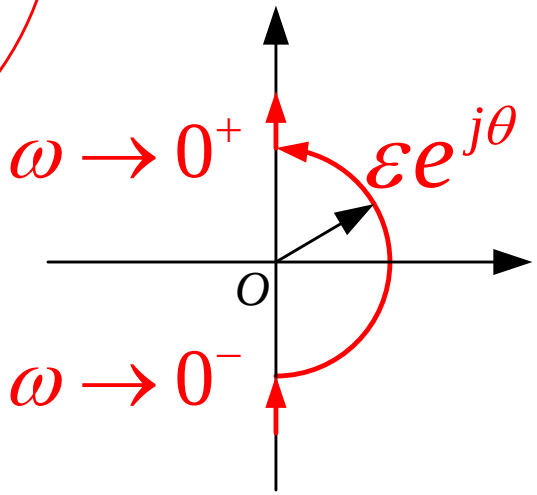
例：某反馈控制系统开环传递函数为

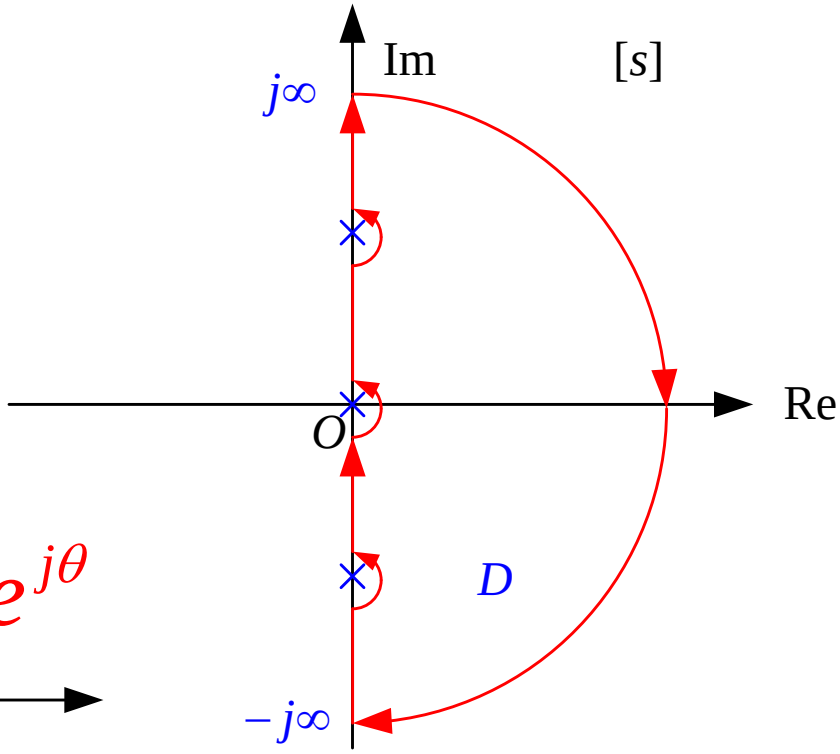
$$G(s)=\frac{K}{s(0.1s+1)(0.05s+1)}$$

判断当K=10和40时的稳定性

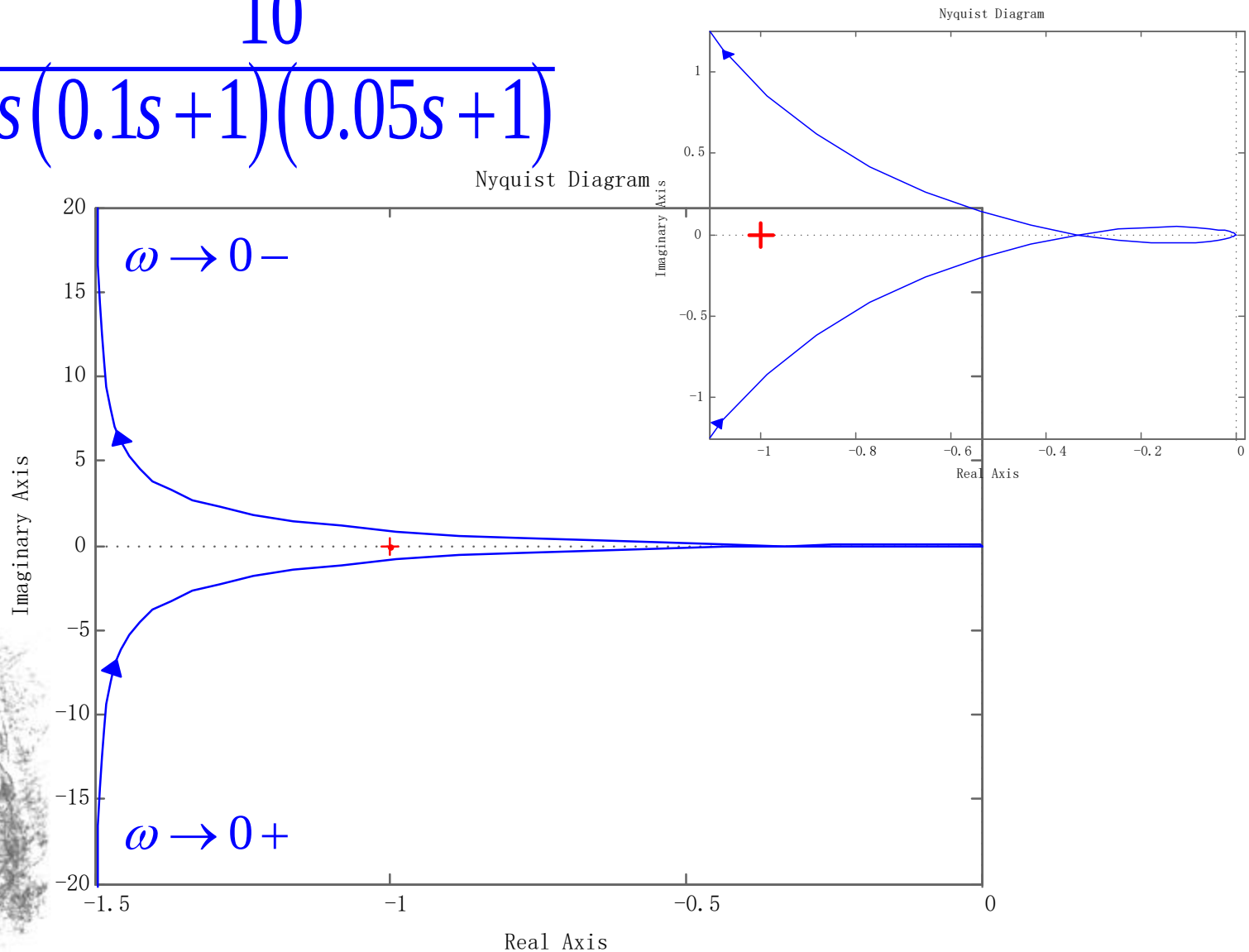
如果开环传递函数在虚轴上有极点或零点，







$$G(s)=\frac{10}{s(0.1s+1)(0.05s+1)}$$

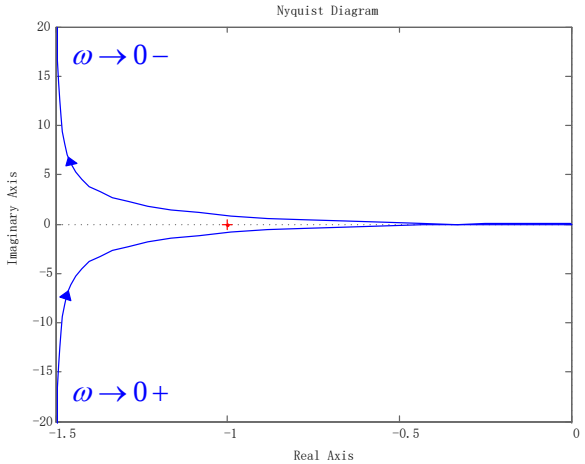


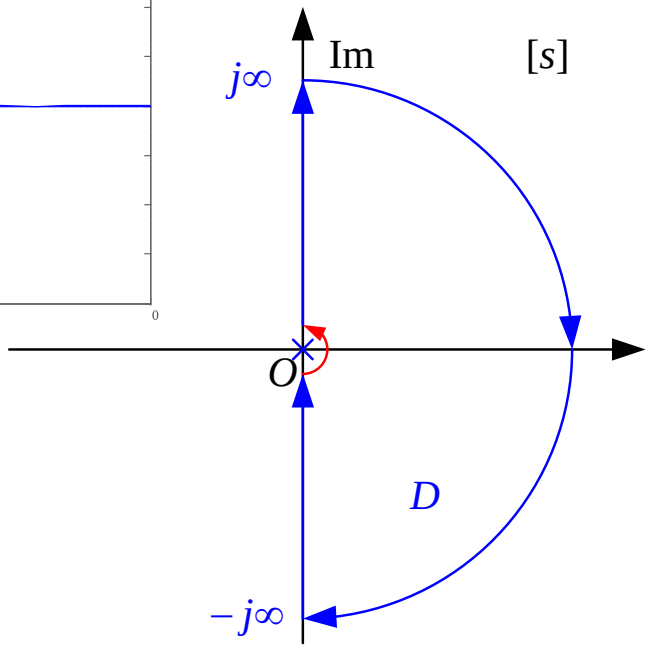
$$s = \varepsilon e^{j\theta}$$

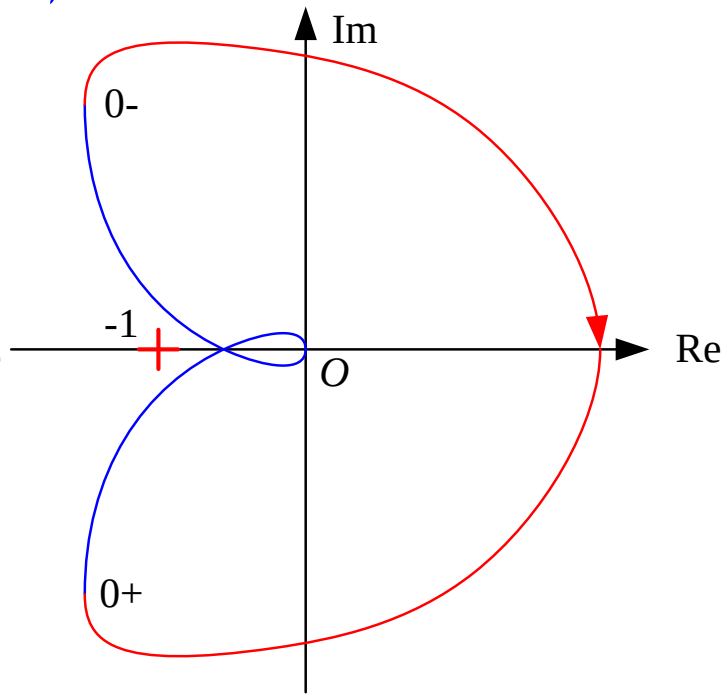
$$\theta = -90^\circ \to 0^\circ \to 90^\circ$$

$$G\left(\varepsilon e^{j\theta}\right) \cong \frac{10}{\varepsilon e^{j\theta}} = +\infty e^{-j\theta}$$

$$\angle G\left(\varepsilon e^{j\theta}\right) = 90^\circ \to 0^\circ \to -90^\circ$$







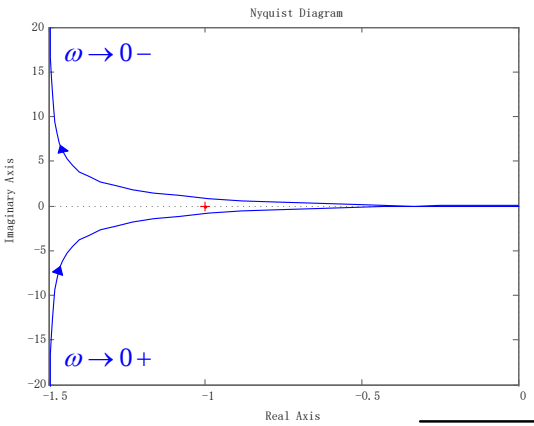
开环没有右极点，乃氏图不包围(−1, j0)，稳定

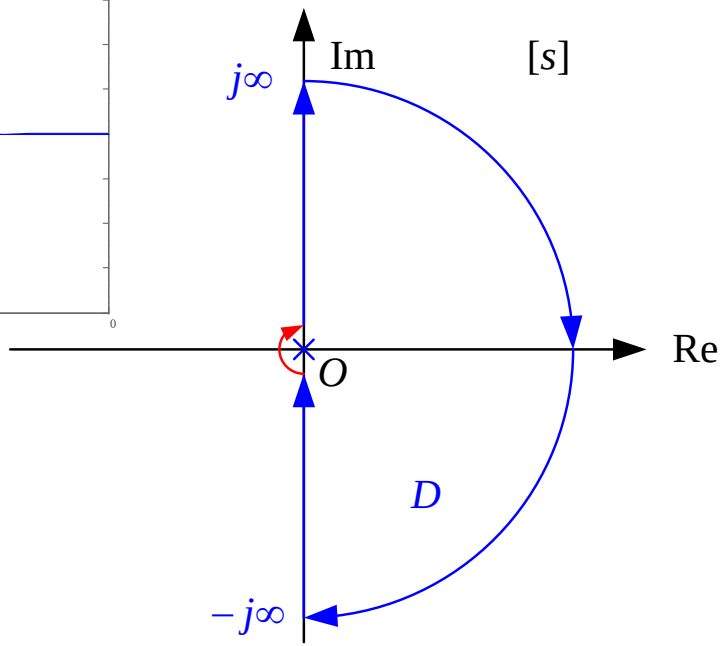
$$s = \varepsilon e^{j\theta}$$

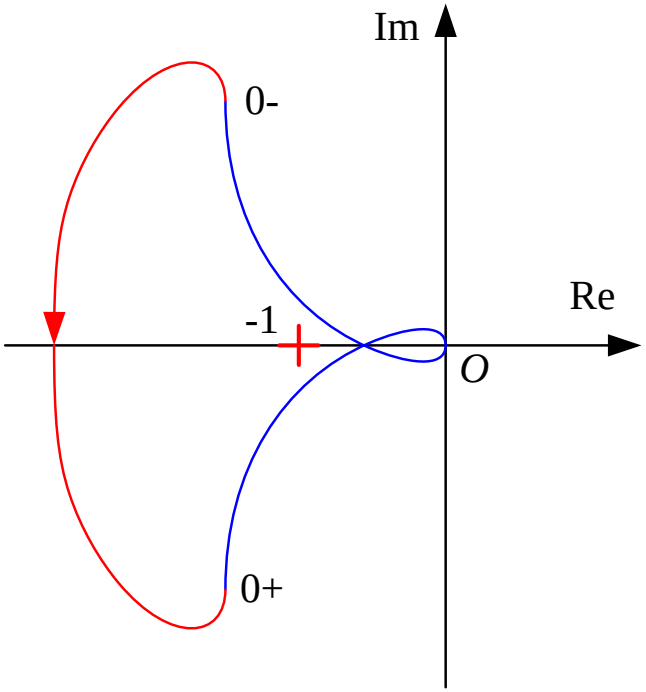
$$\theta = -90^\circ \to -180^\circ \to -270^\circ$$

$$G\left(\varepsilon e^{j\theta}\right) \cong \frac{10}{\varepsilon e^{j\theta}} = +\infty e^{-j\theta}$$

$$\angle G = +90^\circ \to +180^\circ \to +270^\circ$$

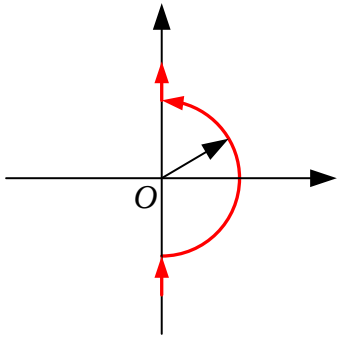


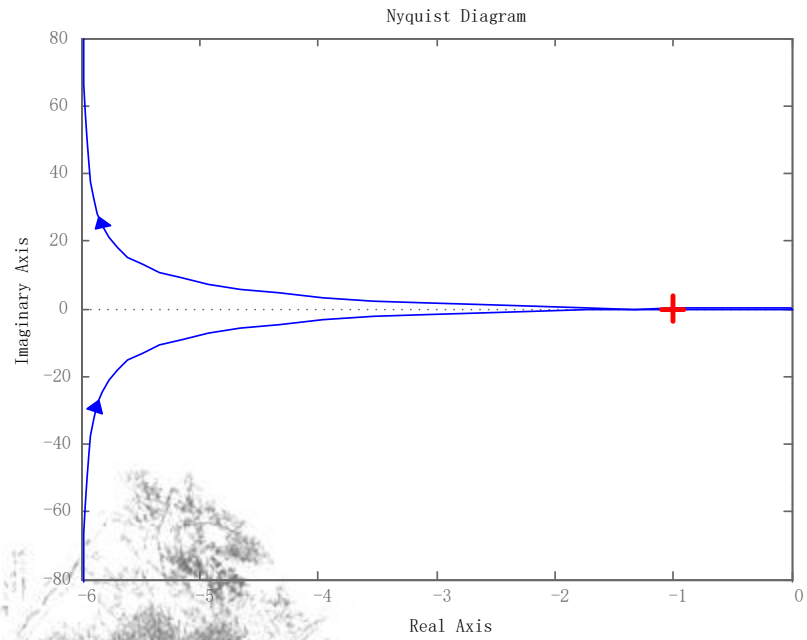


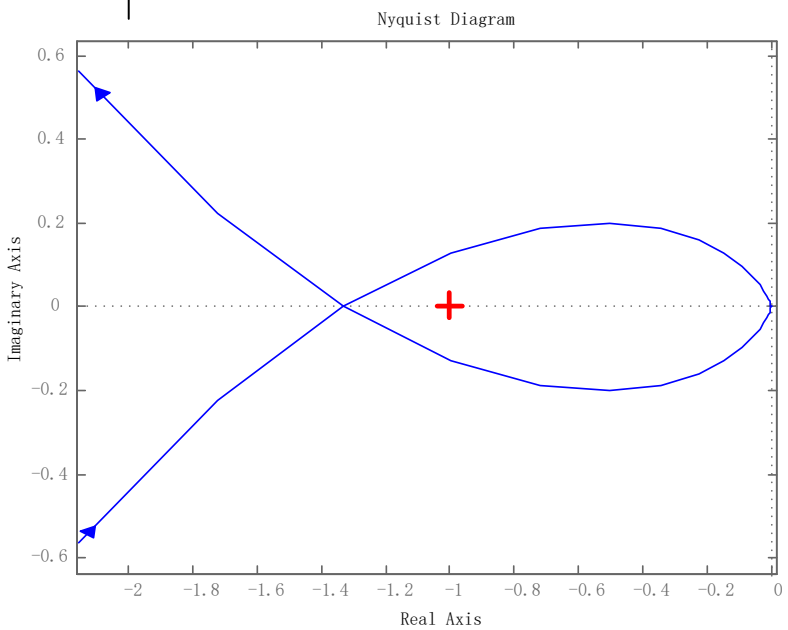


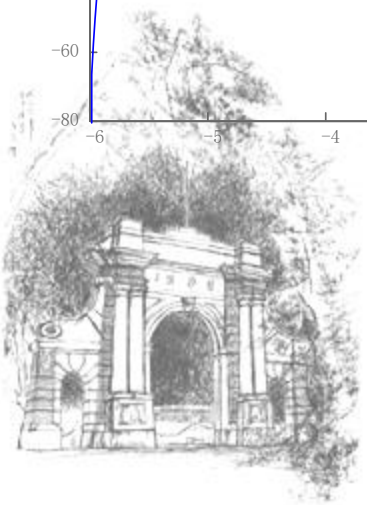
开环右极点有1个，乃氏图逆时针包围$(-1, j0)$ 1圈，稳定

$$G(s)=\frac{40}{s(0.1s+1)(0.05s+1)}$$





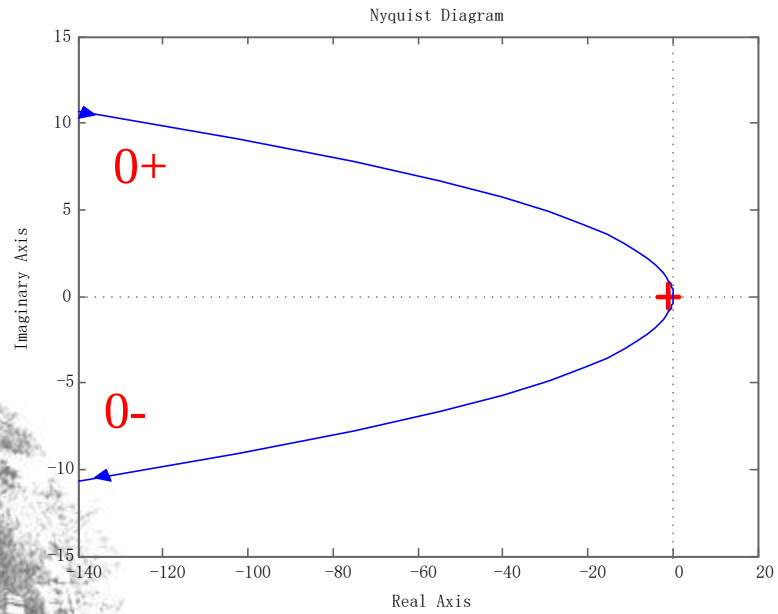


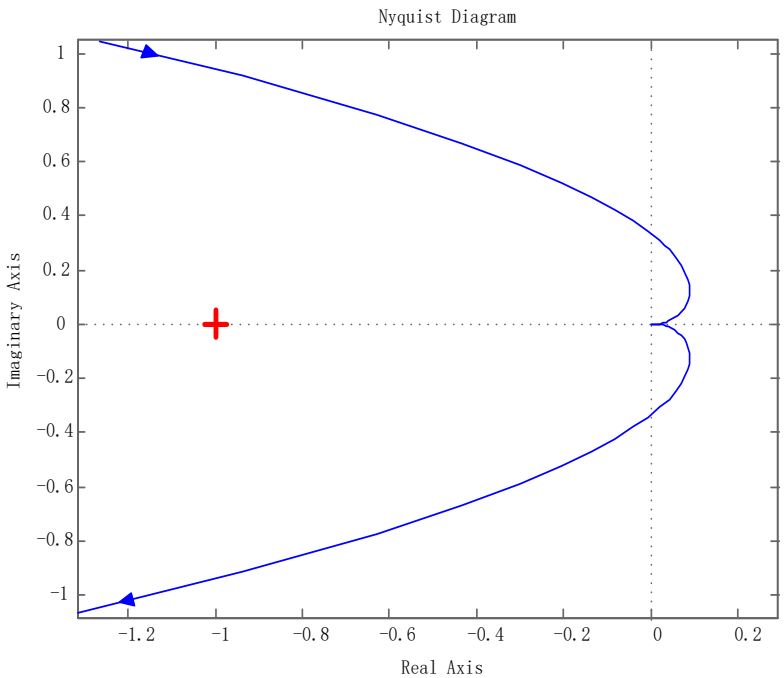


从原点右边绕，开环右极点个数为0；
乃氏图顺时针包围(-1, j0) 2 圈，不稳定

例：某系统开环传递函数为

$$G(s)H(s)=\frac{4(0.05s+1)}{s^2(0.3s+1)(0.05s^2+0.2s+1)}$$





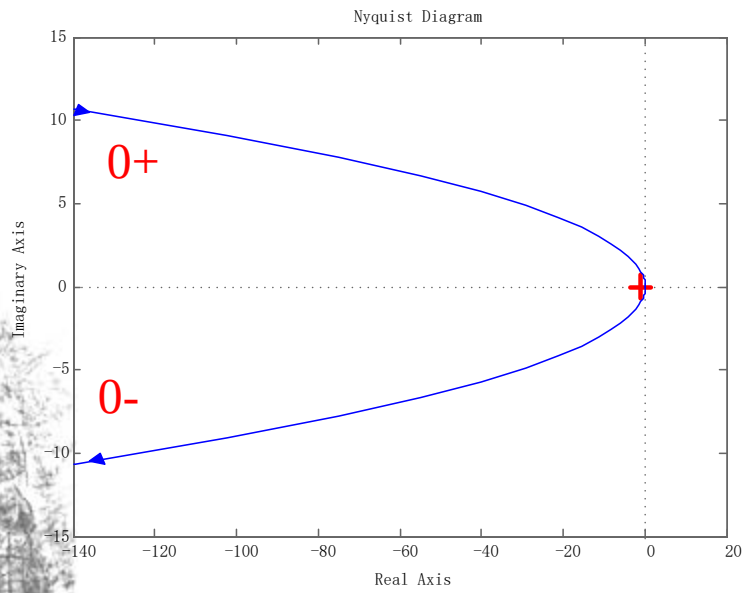
从原点右边绕，$s=\varepsilon e^{j\theta}$

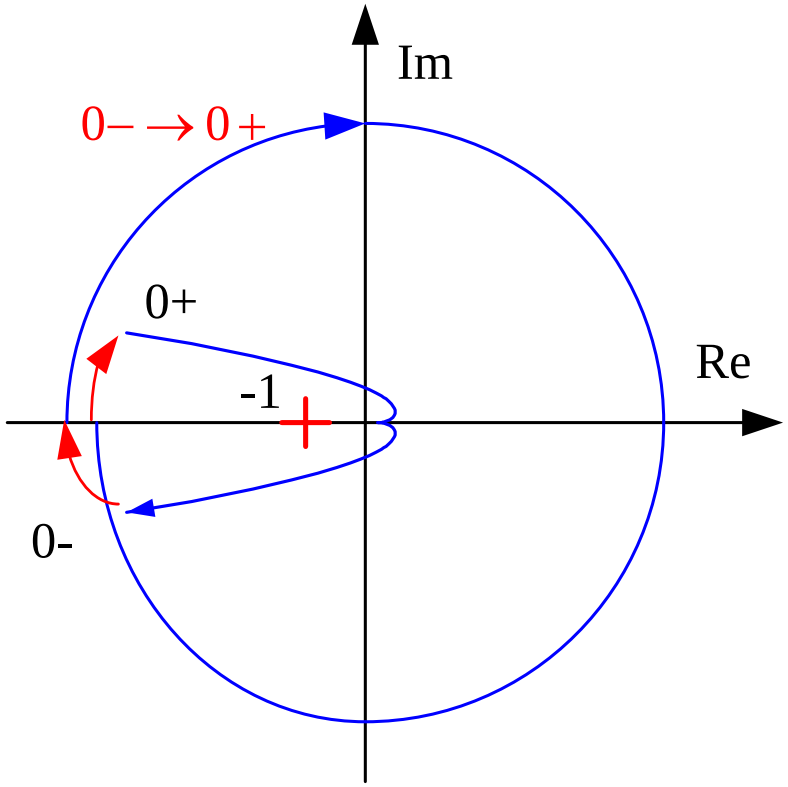
$$\theta=-90^\circ\to 0^\circ\to 90^\circ$$

$$G\left(\varepsilon e^{j\theta}\right)H\left(\varepsilon e^{j\theta}\right)$$

$$\cong\frac{4}{\left(\varepsilon e^{j\theta}\right)^2}=\frac{4}{\varepsilon^2}e^{-j2\theta}$$

$$\angle GH=180^\circ\to 0^\circ\to -180^\circ$$

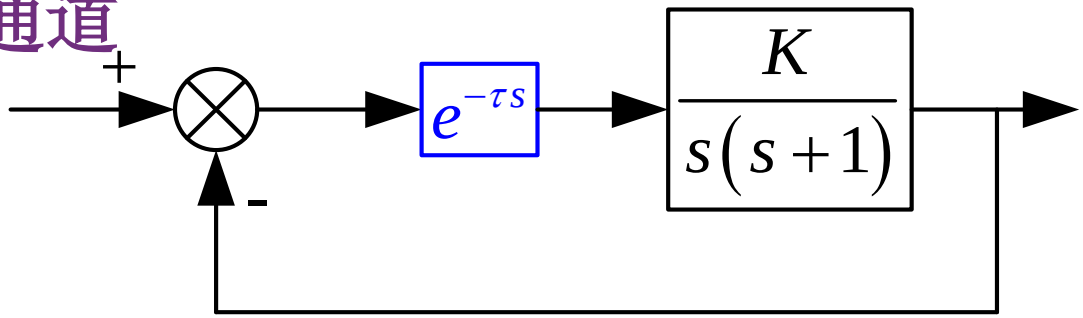




顺时针2圈，不稳定

5.5应用乃奎斯特判据分析延时系统的稳定性

延时环节串联在前向通道



*K*满足什么条件时系统闭环稳定？

$$G_1(j\omega)=\frac{K}{j\omega(j\omega+1)}=\frac{K}{\omega}\cdot\frac{1}{-\omega+j1}$$

$$|G_1|=\frac{K}{\omega}\cdot\frac{1}{\sqrt{\omega^2+1}}\qquad \angle G_1=-180+\arctan\frac{1}{\omega}$$

$$G_2(j\omega)=e^{-j\tau\omega}$$

$$|G_2|=1\qquad \angle|G_2|=-\tau\omega$$

$$|G|=\frac{K}{\omega}\cdot\frac{1}{\sqrt{\omega^2+1}}\qquad \angle G=-180+\arctan\frac{1}{\omega}-\tau\omega$$

$$|G_1| = \frac{K}{\omega} \cdot \frac{1}{\sqrt{\omega^2+1}}$$

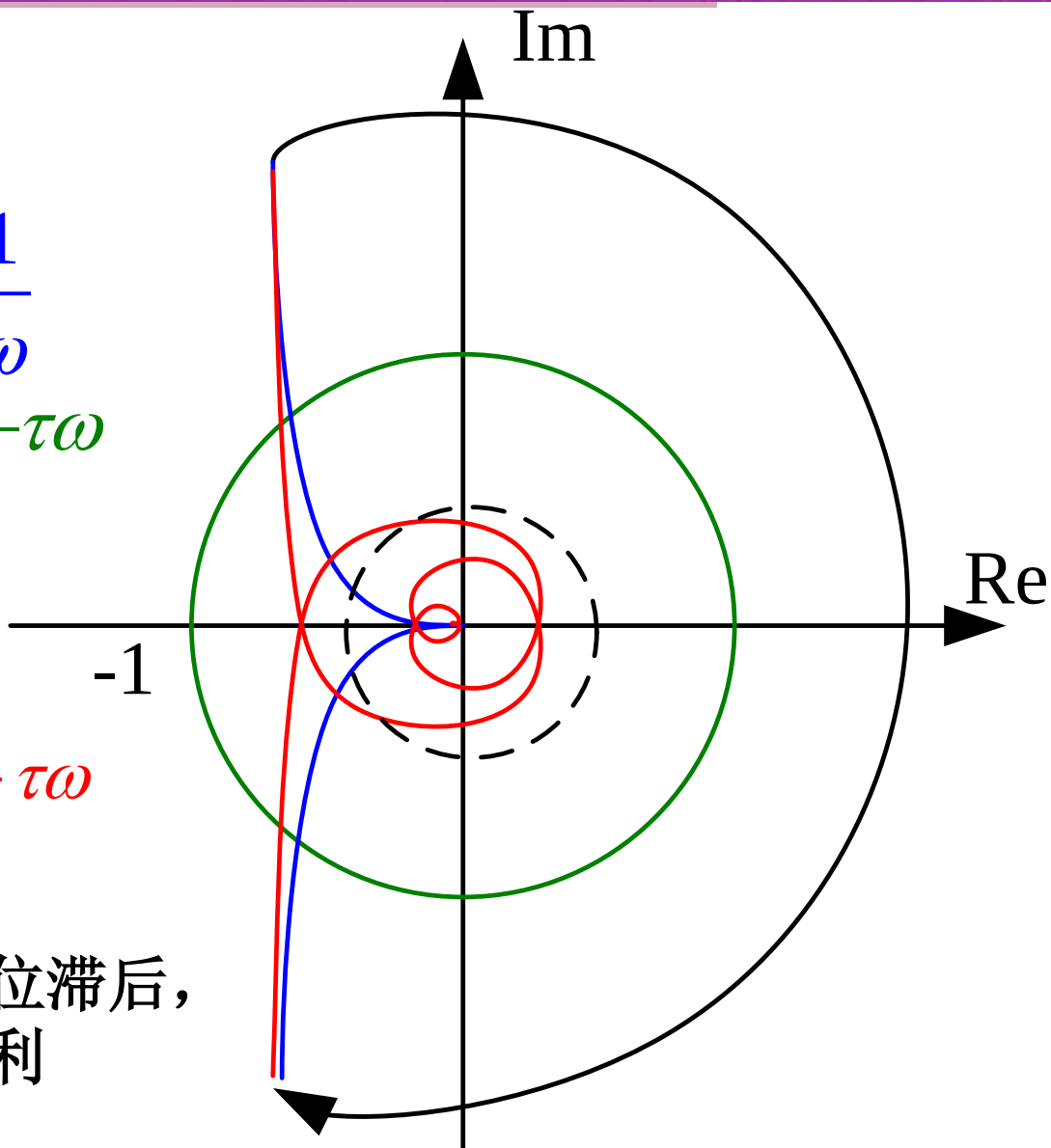
$$\angle G_1 = -180 + \arctan\frac{1}{\omega}$$

$$|G_2| = 1 \qquad \angle|G_2| = -\tau\omega$$

$$|G| = \frac{K}{\omega} \cdot \frac{1}{\sqrt{\omega^2+1}}$$

$$\angle G = -180 + \arctan\frac{1}{\omega} - \tau\omega$$

必然导致相位滞后，
对稳定性不利



$$|G| = \frac{K}{\omega} \cdot \frac{1}{\sqrt{\omega^2+1}} \overset{令}{=} 1$$

$$\omega\sqrt{\omega^2+1} = K$$

$$\angle G = -180 + \arctan\frac{1}{\omega} - \tau\omega > -180$$

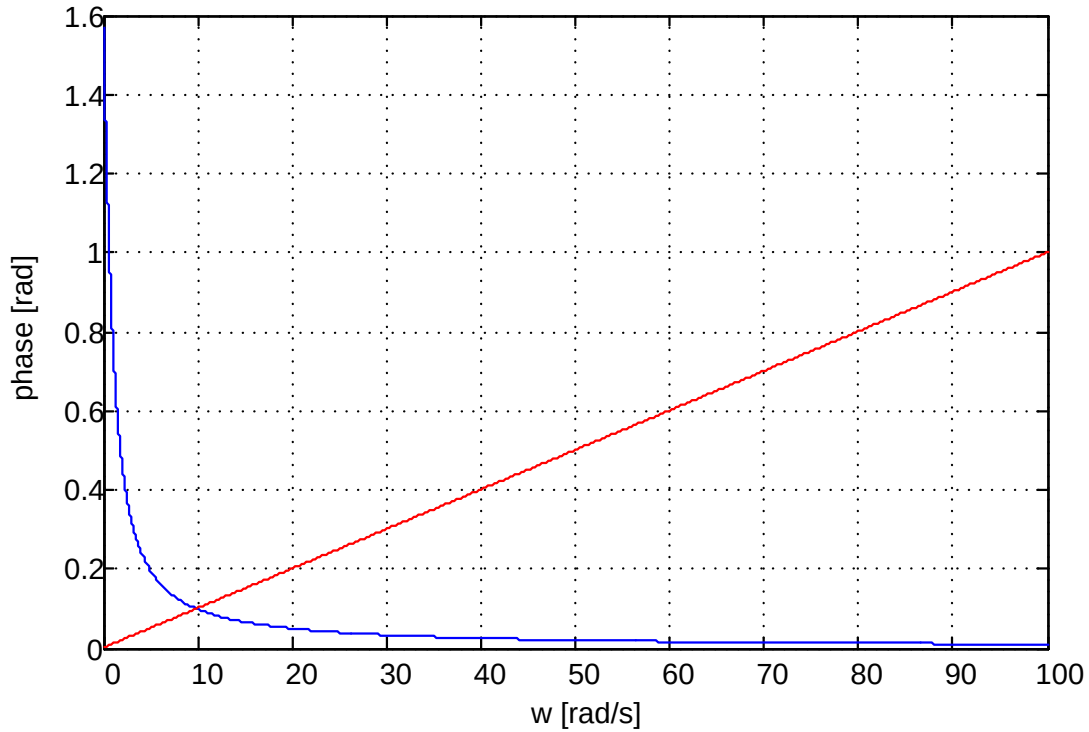
$$\arctan\frac{1}{\omega} > \tau\omega$$

例如 $\tau = 10\ \text{ms}$ $\omega < 10\ \text{rad/s}$ $K < 100$

$\tau = 1\ \text{ms}$

$\omega < 31.6\ \text{rad/s}$

$K < 1000$



Nyquist稳定判据

在[*s*]平面作包围右半平面的*D*形曲线，
如果开环传递函数的Nyquist图逆时针包围(－1, j0)点的圈数等于开环右极点的个数，
则系统稳定。

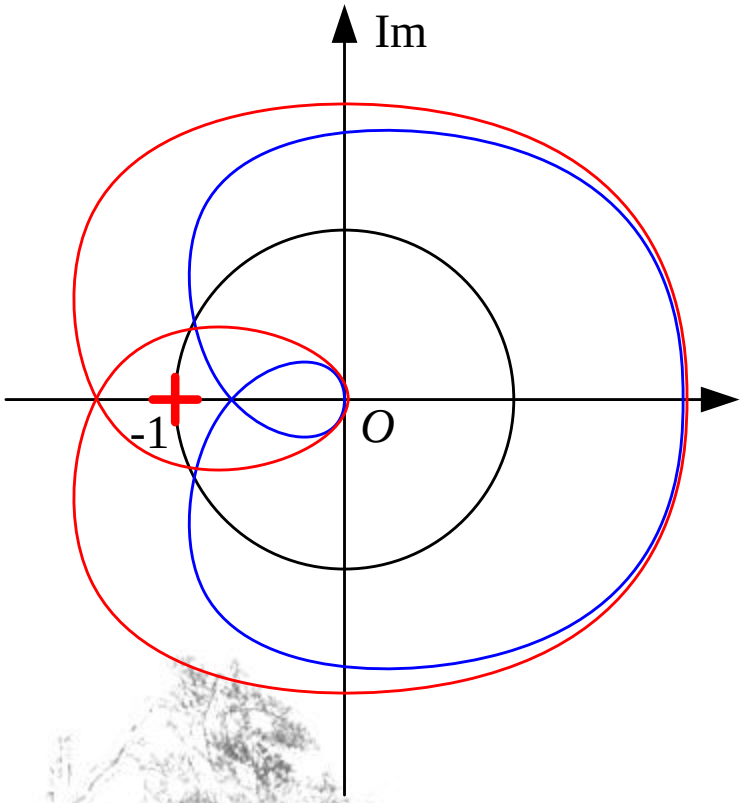
(1) 开环右极点个数如何判断？——特征方程

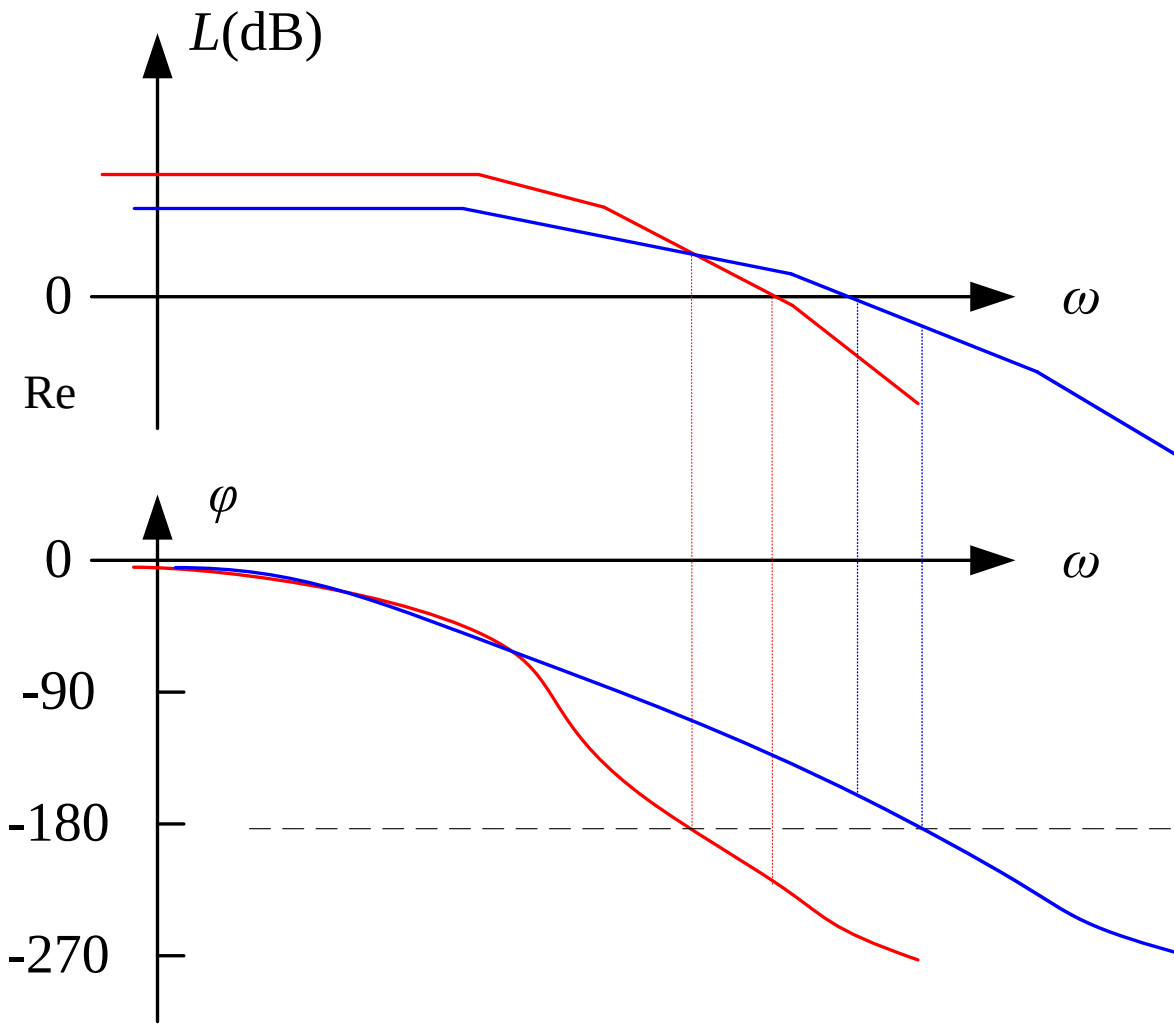
(2) 开环在虚轴上有零极点？——绕道

(3) 开环无右极点——不包围

(4) 乃氏判据也适用于有延时环节的情况

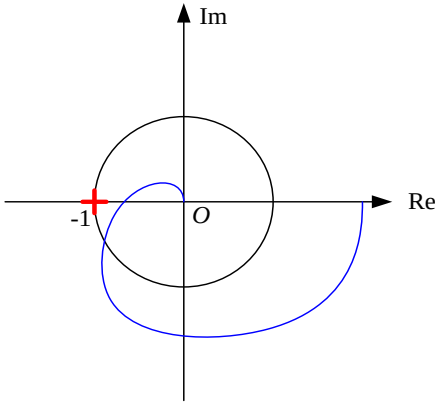
5.6 由伯德图判断系统的稳定性



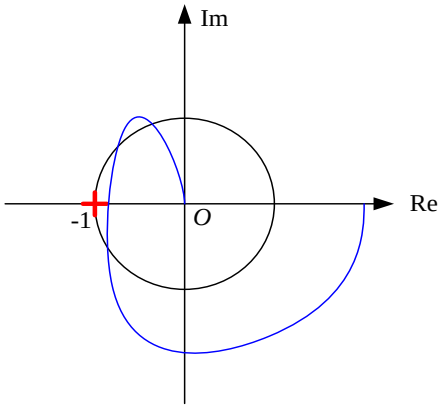


单位圆→0dB线，与实轴负半轴的交点-180°

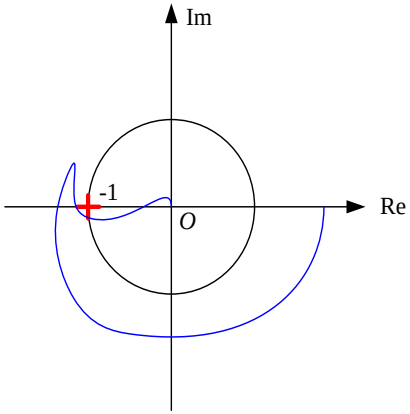
稳定



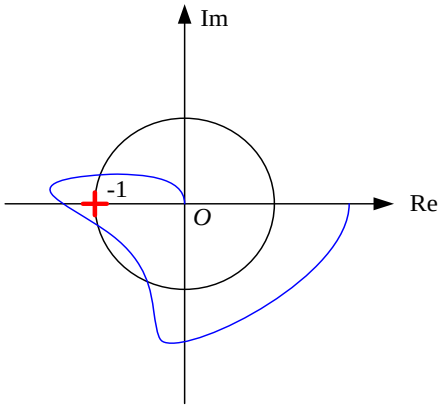
稳定



不稳定



不稳定



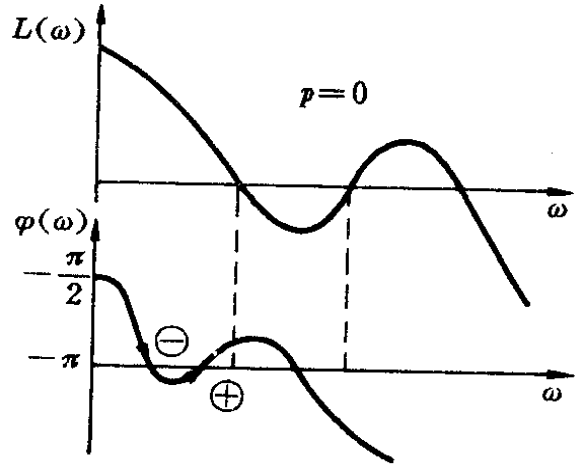
开环特征多项式没有位于右半平面的根，且在L(w)≥0 dB的范围内，相位都大于-180°，那么系统稳定。

由伯德图判稳定性

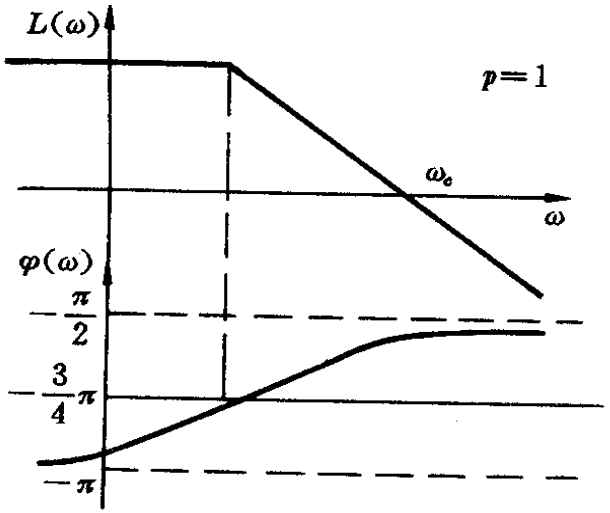
设0型或I型系统开环特征方程有 p 个右根，
且开环静态放大倍数大于零，
如果在所有$L(\omega)\geq 0$频率范围内，
相频特性曲线$\varphi(\omega)$在$(-\pi)$线上正负穿越之差为$p/2$次，
则闭环系统稳定。

乃氏图从第三象限穿越负实轴到第二象限，负穿越；
从第二象限穿越负实轴到第三象限，正穿越。

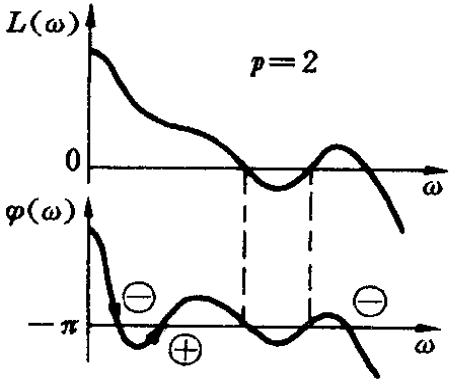
如果$\omega=0$时，$\varphi(\omega)=-\pi$，
乃氏图向第三象限去，半次正穿越，
向第二象限去，半次负穿越。



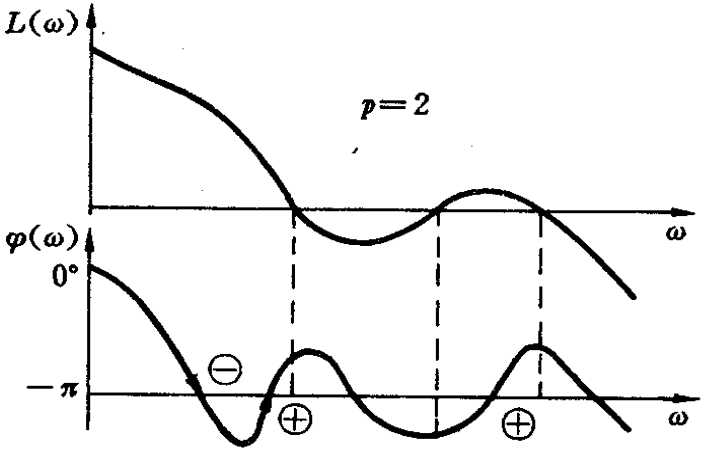
(a)



(b)



(c)



(d)

图（a），已知$p=0$，即开环无右特征根，在$L(\omega)>0$范围内，正负穿越之差为0，系统闭环稳定。

图（b），已知开环传递函数有一个右极点，$p=1$，在$L(\omega)>0$的频率范围内，半次正穿越，系统闭环稳定。

图（c），已知$p=2$，在$L(\omega)>0$的范围内，正负穿越之差为$1-2=-1\neq 2/2$，系统闭环不稳定。

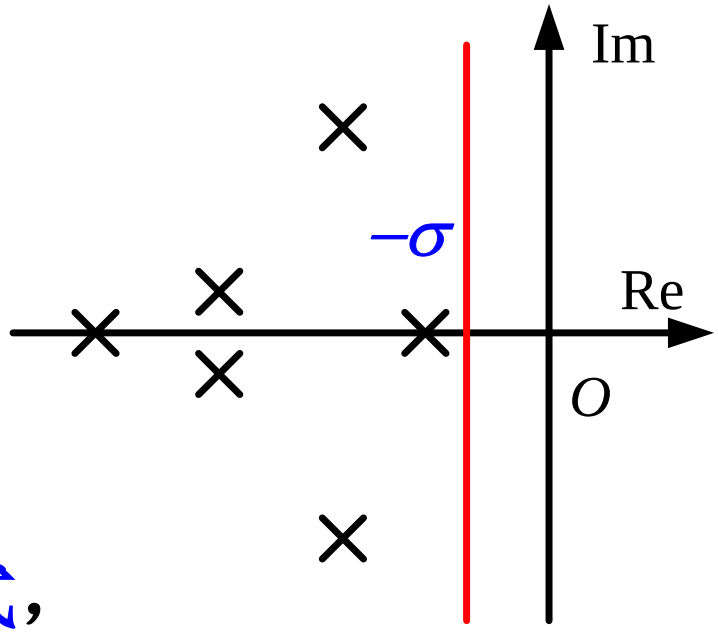
图（d），已知$p=2$，在$L(\omega)>0$的范围内，正负穿越之差为$2-1=1=2/2$，系统闭环稳定。

5.7控制系统的相对稳定性

定性→定量

用劳斯判据

如果系统闭环特征根均在s左半平面，且和虚轴有一段距离，则系统有一定的稳定裕量。



虚轴左移σ，令$z=s+\sigma$，将$s=z-\sigma$代入系统特征式，得到z的方程式，采用劳斯判据，可知距离虚轴σ以右是否有根。

例：
$$\frac{X_o(s)}{X_i(s)}=\frac{10000(0.3s+1)}{s^4+10s^3+35s^2+50s+24}$$

令 $z = s+1$，即 $s = z - 1$，代入系统特征式，

$$(z-1)^4+10(z-1)^3+35(z-1)^2+50(z-1)+24=0$$

即 $$z^4+6z^3+11z^2+6z=0$$

z^4	1	11	0
z^3	6	6	
z^2	10	0	
z^1	6		
z^0	1		

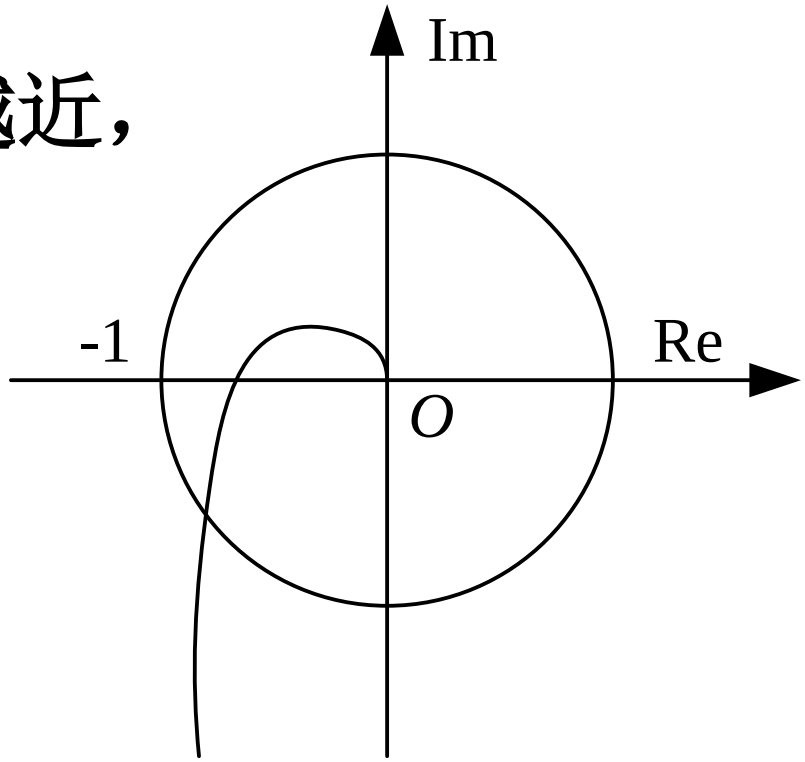
z的多项式系数无相反符号，劳斯阵列第一列未变号，系统在s=-1以右没有根。

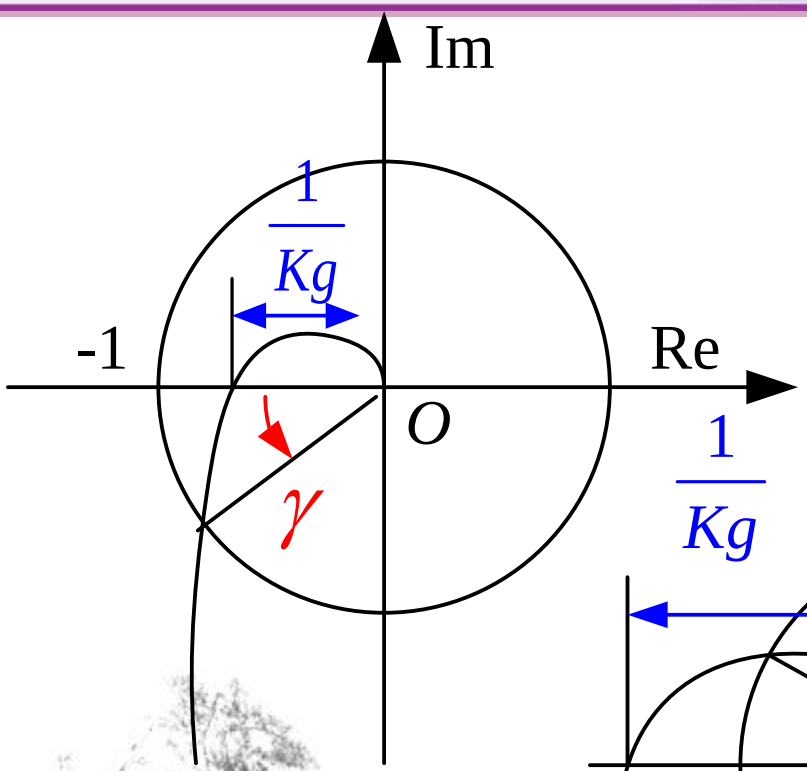
实际4个根为-1, -2, -3, -4

用Nyquist图

如果系统稳定，
Nyquist图离 (-1, j0) 越近，

相对稳定性越差。

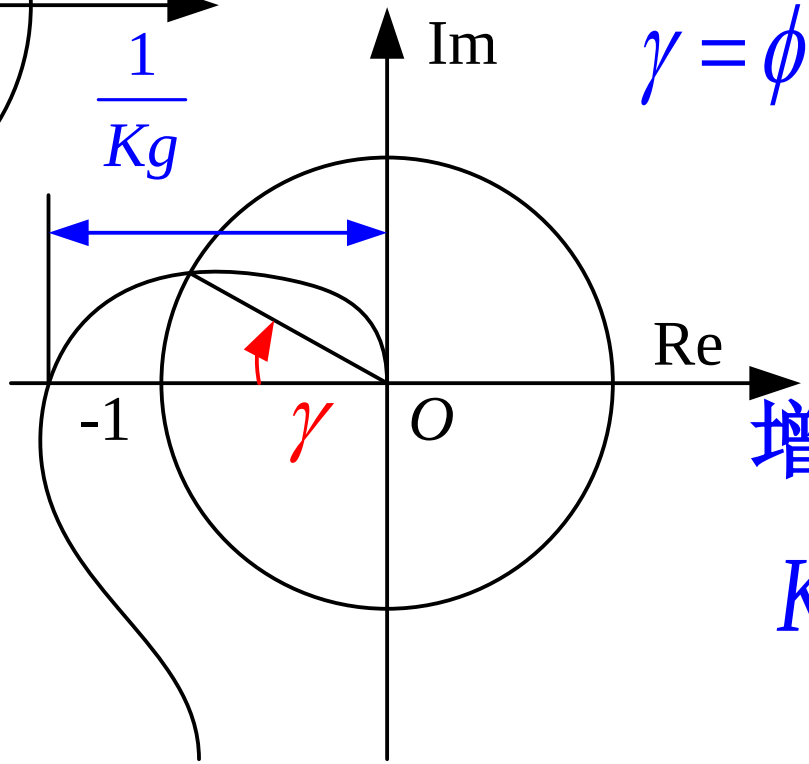




$$\left|G\left(j\omega_c\right)\right|=1$$ 剪切频率

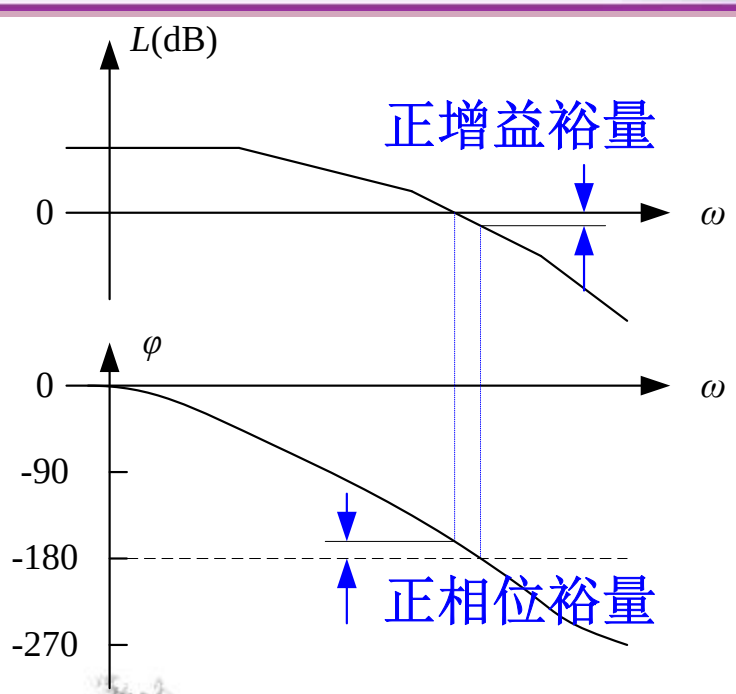
相位裕量

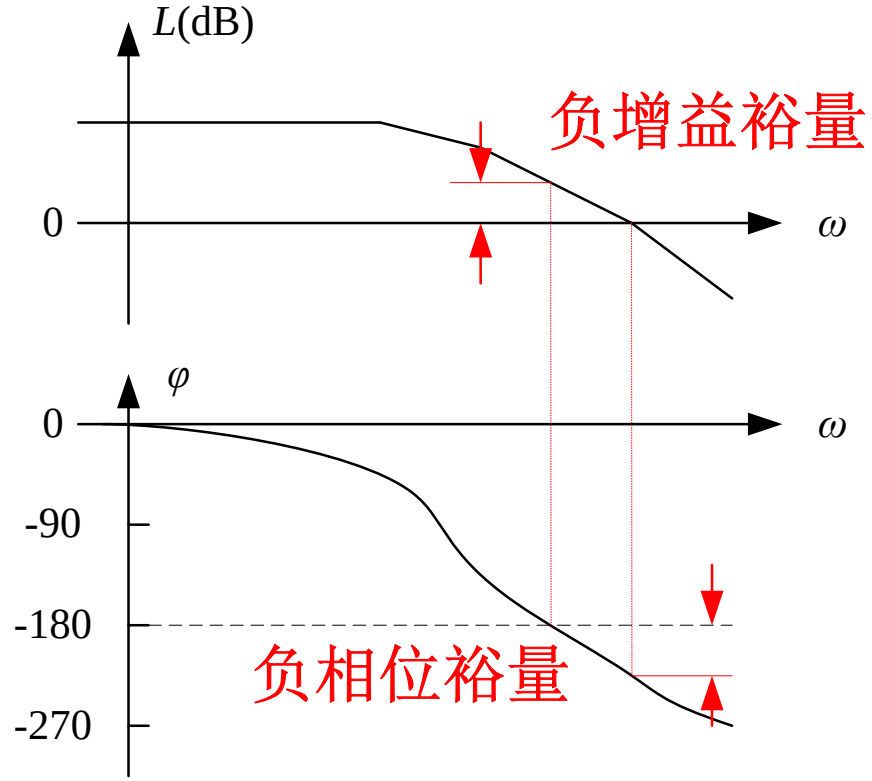
$$\gamma=\phi\left(\omega_c\right)-\left(-180^\circ\right)$$



增益裕量

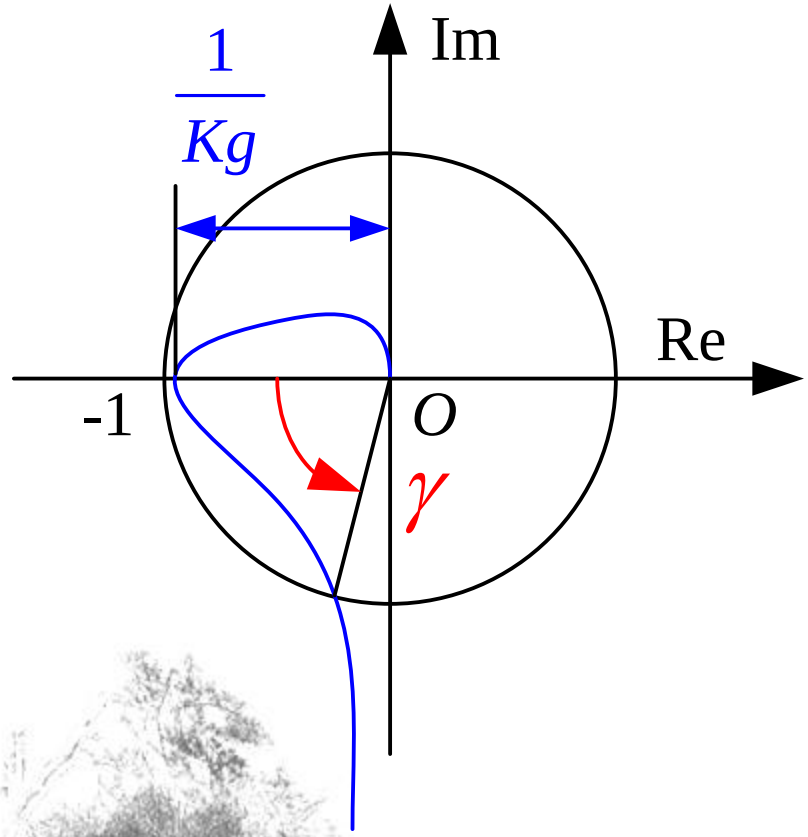
$$K_g=\left|\frac{1}{G\left(j\omega_{-\pi}\right)}\right|$$

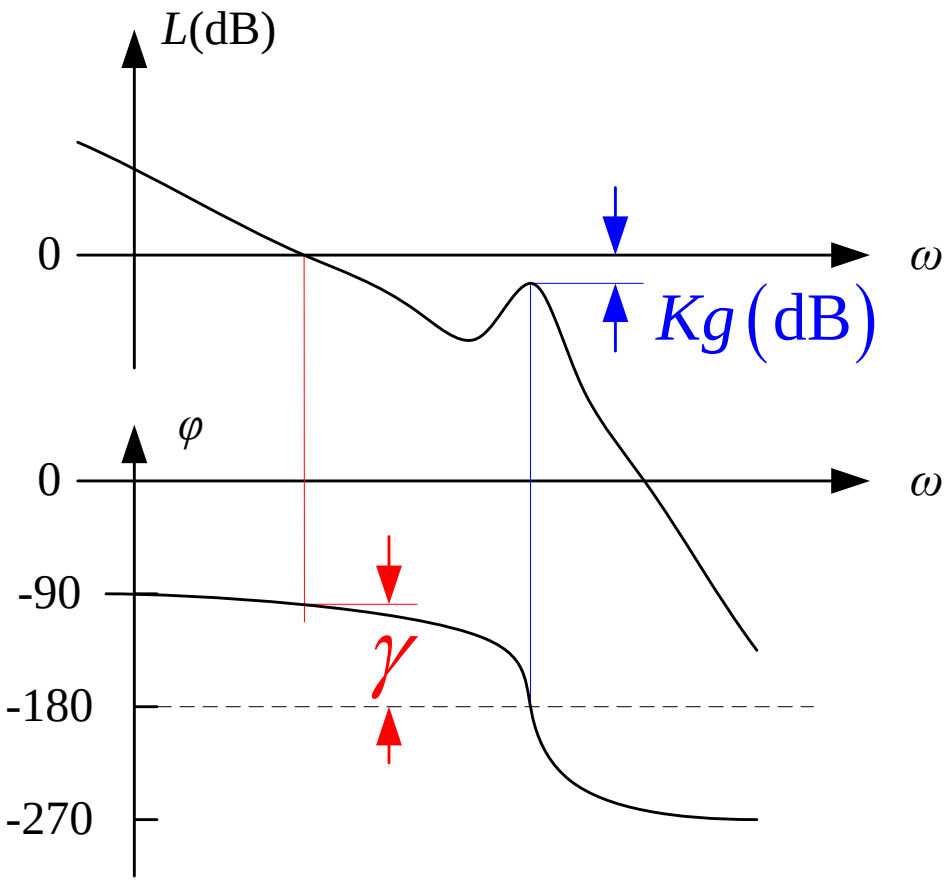




增益裕量也可用分贝数表示：

$$20\lg K_g = 20\lg\left|\frac{1}{G\left(j\omega_{-\pi}\right)}\right| = -20\lg\left|G\left(j\omega_{-\pi}\right)\right|$$

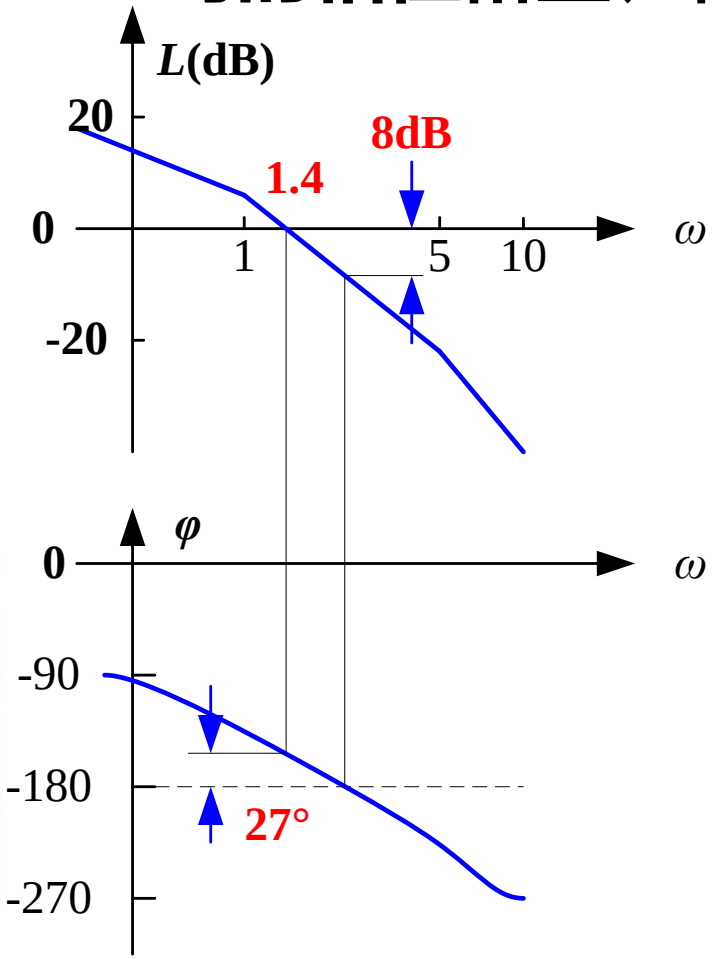

Im
$\frac{1}{Kg}$
Re
-1
O
γ


L(dB)
0
ω
Kg(dB)
φ
0
ω
-90
γ
-180
-270

例：开环传递函数为 $G(s)=\dfrac{K}{s(s+1)(s+5)}$

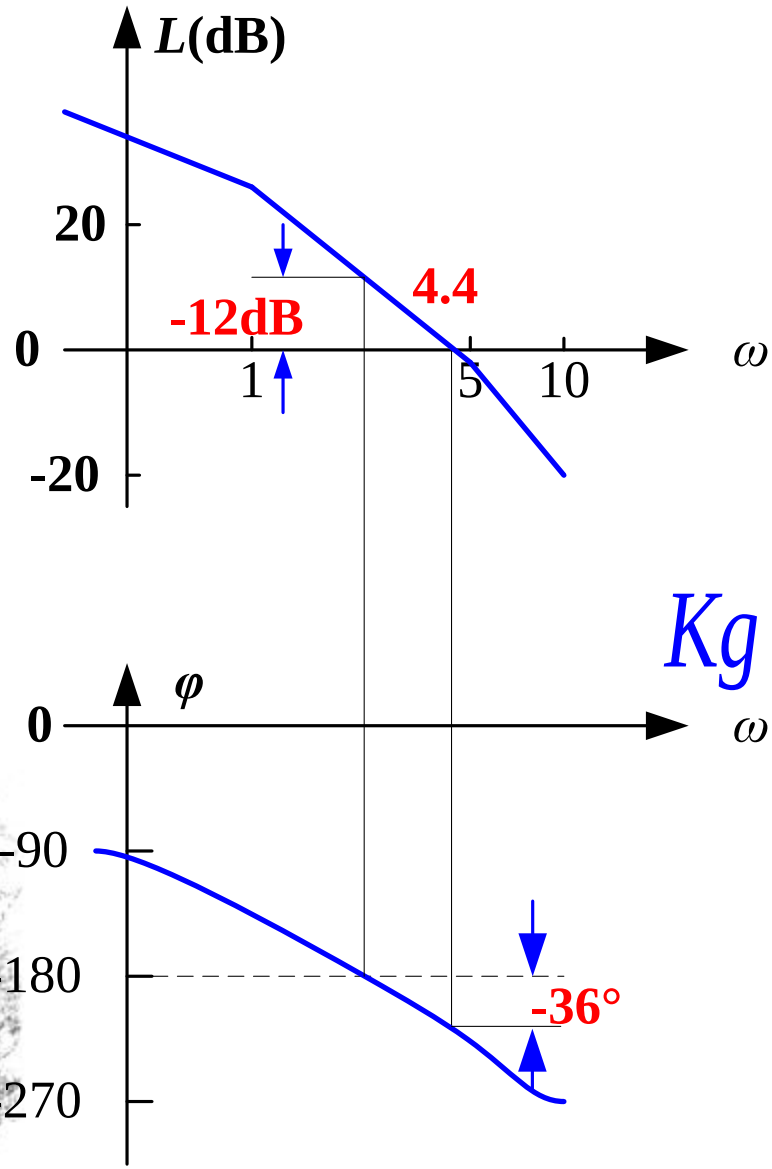
求K=10、100时的相位裕量、幅值裕量。

K=10



	作图法	计算法
ω_c	≈ 1.4	1.23 rad/s
γ	$27°$	$25.5°$
Kg	8dB	9.5dB

K=100



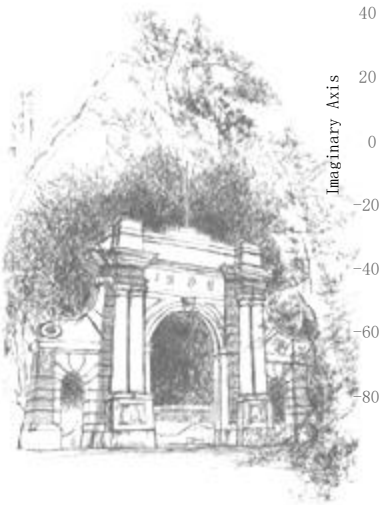
	作图法	计算法
ω_c	≈ 4.4	3.92 rad/s
γ	$-36°$	$-23.5°$
Kg	-12dB	-10.5dB

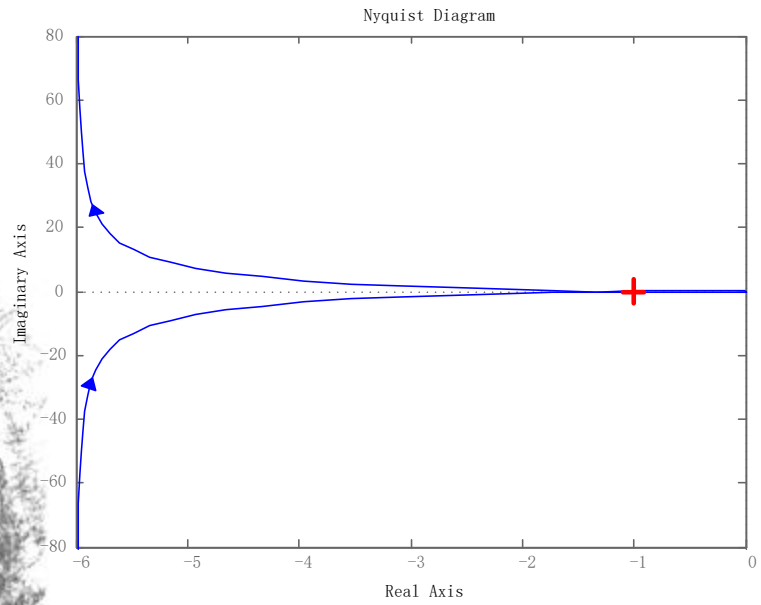
相关matlab函数

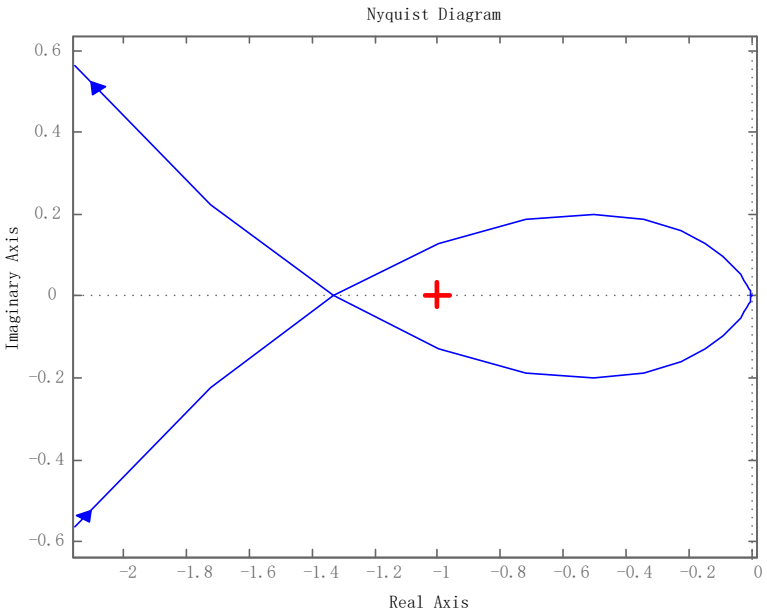
```
s1 = tf([40],[0.005 0.15 1 0])
s1 = zpk([],[0 -10 -20],8000)
nyquist(s1)
```

$$G(s)=\frac{40}{s(0.1s+1)(0.05s+1)}$$

$$G(s)=\frac{8000}{s(s+10)(s+20)}$$







求稳定裕量

[Gm,Pm,Wcg,Wcp] = margin(s1)

```
Warning: The closed-loop system is unstable.
> In lti.margin at 89
Gm =
  7.5000e-001
Pm =
 -7.5156e+000
Wcg =
  1.4142e+001
Wcp =
  1.6259e+001
```

Gm —— 增益裕量

Pm —— 相位裕量

Wcg —— 相位-π处的频率

Wcp —— 剪切频率